Statistics

~12 contact hours

CHAPTERS

- Understand what statistics is and why it matters in engineering and management.
- Learn to distinguish between correlation and causation.
- Explore the concept of random variables and probability functions.
- Descriptive Vs. Inferential Statistics
- Hypothesis Tests
- Confidence intervals

WHY THIS MODULE MATTERS

Provides the statistics basics for data science. Do you have enough data for you anlysis? Are results robust or they happened by chance?

What Is Statistics?

A branch of mathematics dealing with the **collection**, **analysis**, **interpretation**, and **presentation** of masses of numerical data”

Merriam-Webster dictionary

Why It Matters in Practice

- Make data-driven decisions in operations, finance, and strategy.
- Model uncertainty in supply chains, forecasting, and risk analysis.
- Evaluate performance of systems and processes using measurable indicators.
- Support innovation through experimentation and hypothesis testing.

Examples

- Forecasting demand for products using historical sales data.
- Evaluating performance of different supply chain strategies.
- Analyzing customer behavior to improve marketing campaigns.
- Assessing risk in investment or project planning.
- Optimizing resource allocation based on statistical models.
- Compute the probability of events taking place (e.g., failures)

Recap: Random Variable

A random variable assigns a numerical value to each possible outcome of a random experiment. **The probability of a value is the probability of the outcomes that produce it**

Key Idea

- It **assigns numbers** to outcomes in a sample space.
- It allows us to **quantify uncertainty** and analyze patterns in data.

Examples

- Tossing a coin 3 times:
  → Random variable = number of heads (0, 1, 2, or 3)
- Rolling two dice:
  → Random variable = sum of the dice (2 to 12)
- Manufacturing 10 items:
  → Random variable = number of defective items (0,… 10)

Recap: Random Variable - Probability

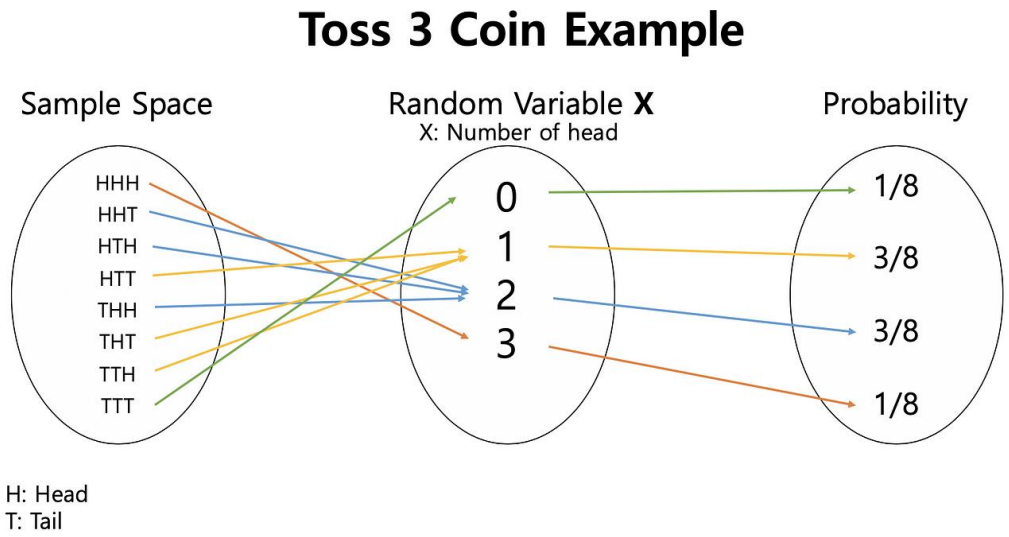
- A random variable assigns a numerical value to each possible outcome of a random experiment. **The probability of a value is the probability of the outcomes that produce it**

Experiment

Toss a coin **three times**.

Each outcome is a sequence like:

- HHH → 3 heads
- HHT → 2 heads
- TTH → 1 head
- TTT → 0 heads

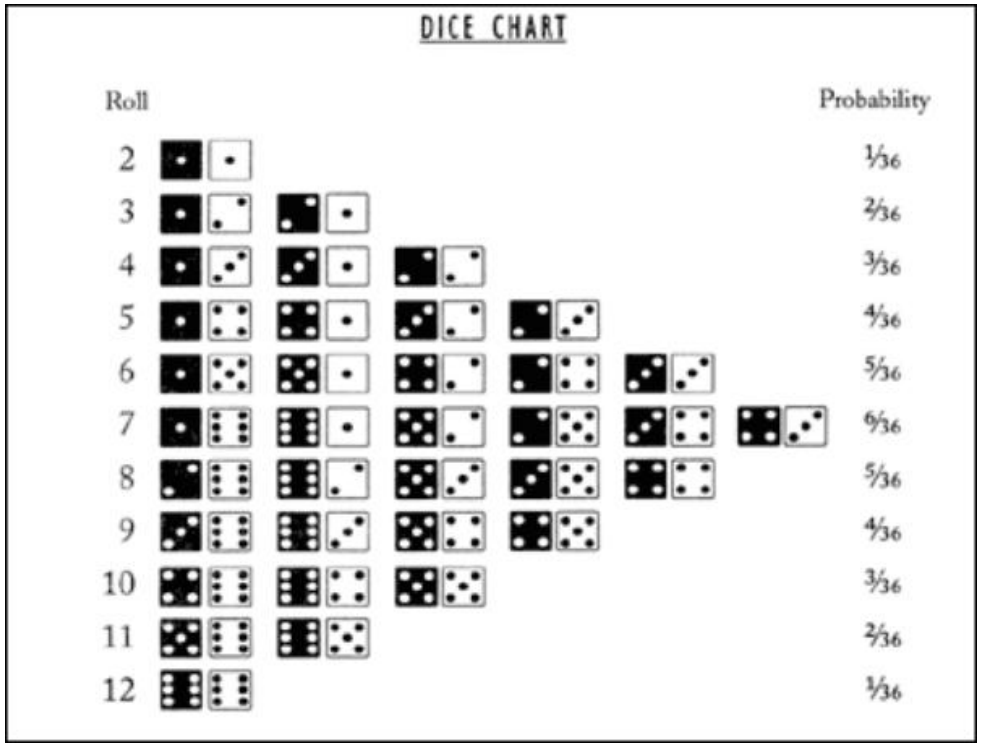


Random Variable: Sum of Two Dice

Sample Space and Values

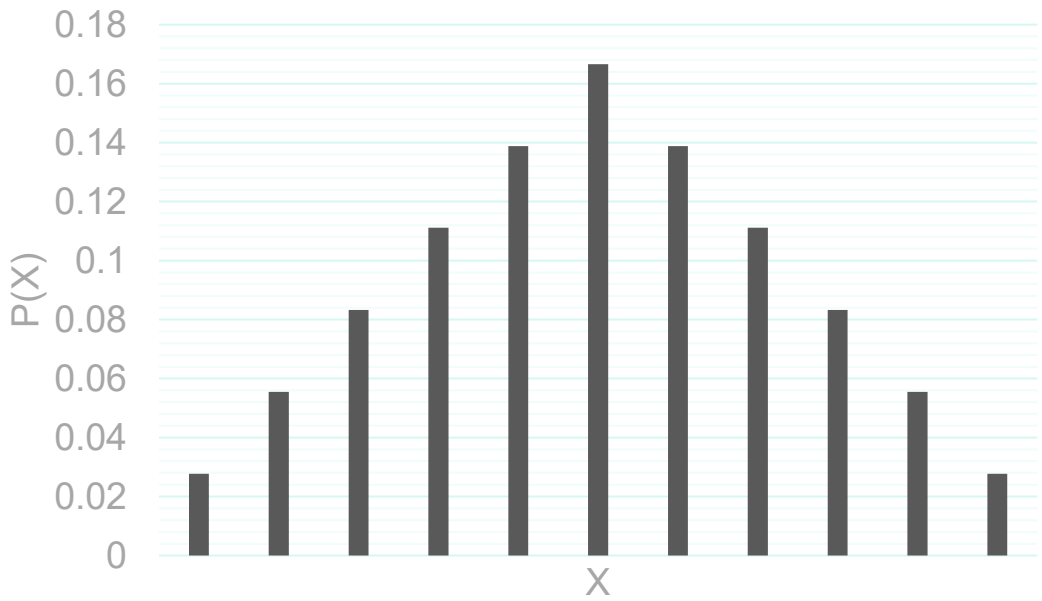
Each outcome (e.g., (1,1), (2,5), (6,6)) maps to a value of X:

- (1,1) → X = 2
- (3,4) → X = 7
- (6,6) → X = 12



Probability Function

X	Probability P(X)
0	0.02
1	0.05
2	0.08
3	0.11
4	0.13
5	0.16
6	0.13
7	0.11
8	0.08
0	0.05
10	0.02



For a discrete random variable X, the probability of a value x, denoted by

$$P(X = x),$$

is the theoretical value toward which the **relative frequency** of observing x tends as the experiment is repeated many times:

$$P(X = x) \approx \frac{\text{number of times } X = x}{\text{total number of trials}}.$$

More precisely, the observed relative frequency is an empirical estimate of $P(X = x)$.

Beware that it is a sample in the case of our example

Continuous Random Variable

Example: standard normal distribution N(0, 1)

Definition

A continuous random variable X can take infinitely many values in an interval. Probabilities are assigned to intervals, not to single exact values.

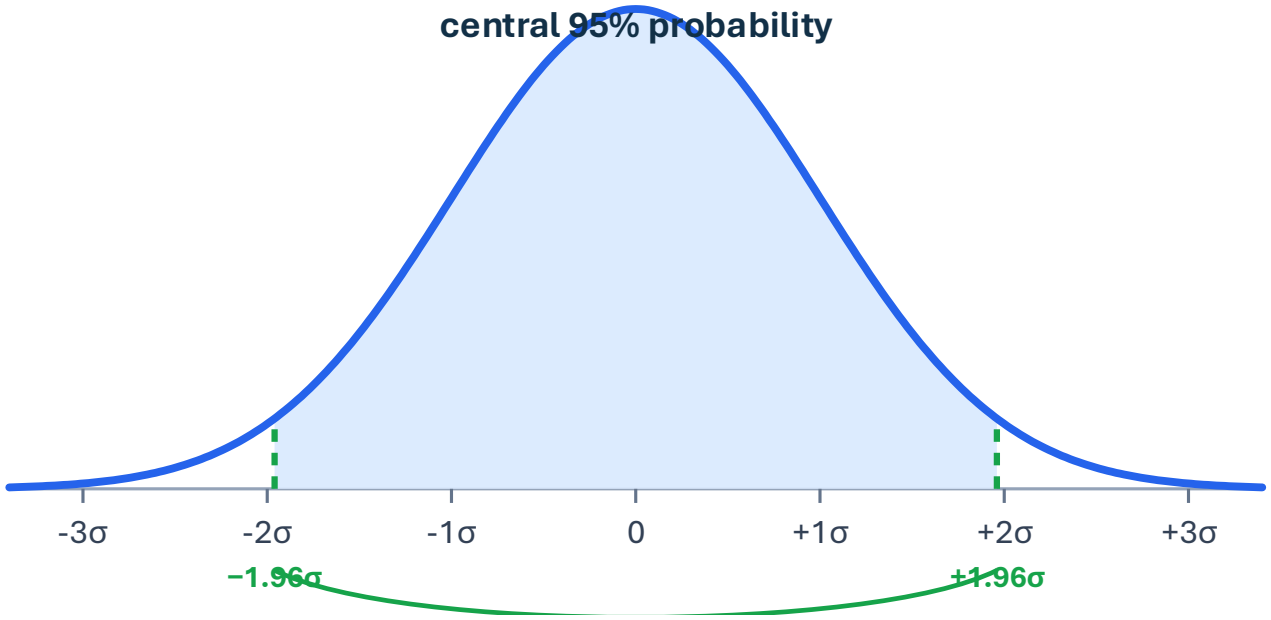
Key idea

$$P(a \le X \le b) = \int_a^b f(x)\,dx$$

For an exact point:

$$P(X = x) = 0$$

but an interval can have positive probability.



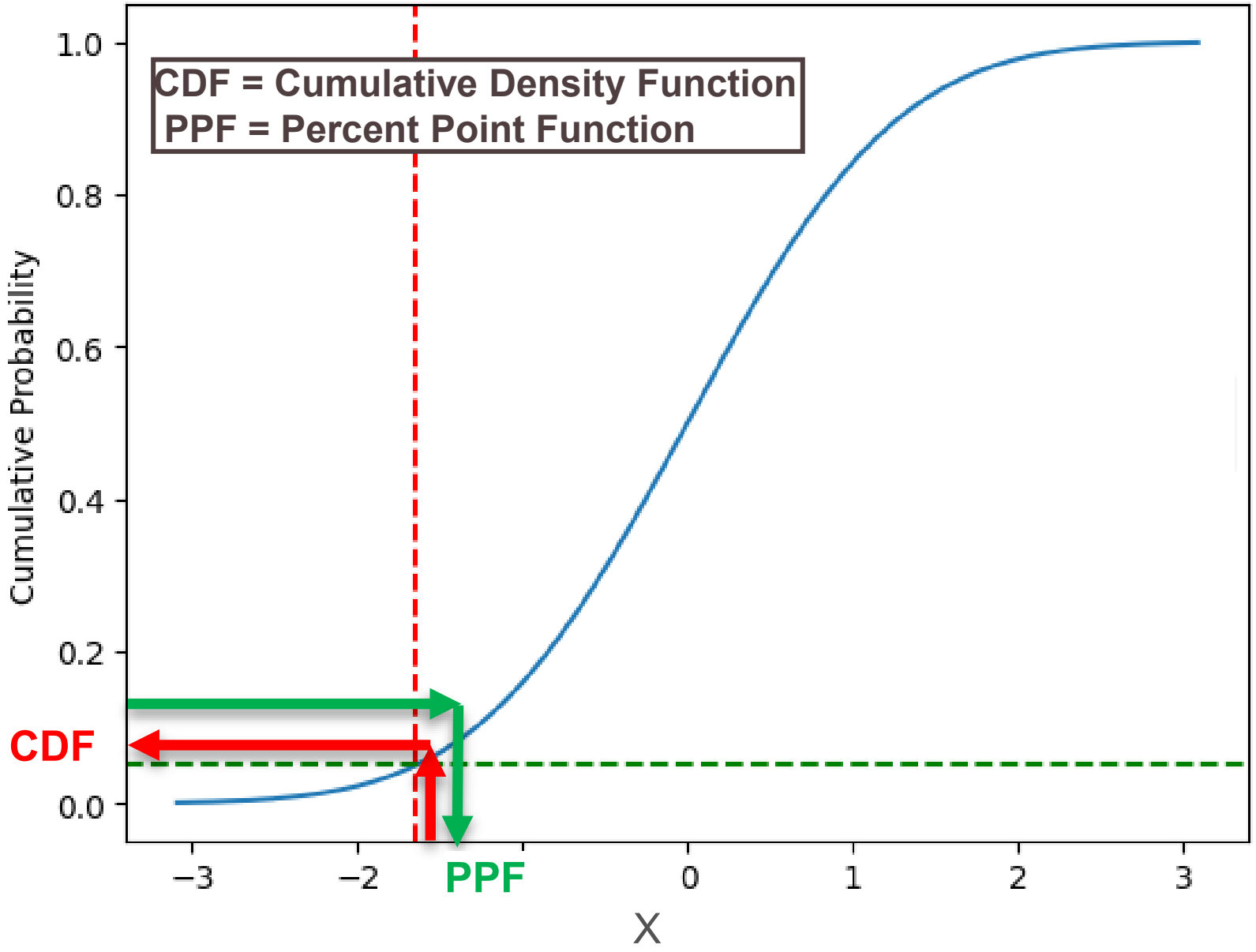
For the standard normal random variable

$$X \sim N(0,1), \quad \mu = 0,\ \sigma = 1$$

The central 95% interval is approximately:

$$[-1.96,\ 1.96]$$

Continuous Random Variable – CDF and PPF



Exercise: defects in a production line

Use the first four probability operations: complement, sum, product, and conditional probability.

Scenario

A factory produces mechanical components. Let D = dimensional defect and S = surface defect.

Given quality-control data, 100 measurements

#(D) = 6 #(S) = 4 #(D and S) = 2

Production line — Quality check — Parts classified — Defect data

Questions

1 Complement

Find $P(D^c)$

What is the probability that a part has no dimensional defect?

2 Sum / OR

Find $P(D \cup S)$

What is the probability that a part has at least one of the two defects?

3 Product / AND

Find $P(D_1 \cap D_2)$

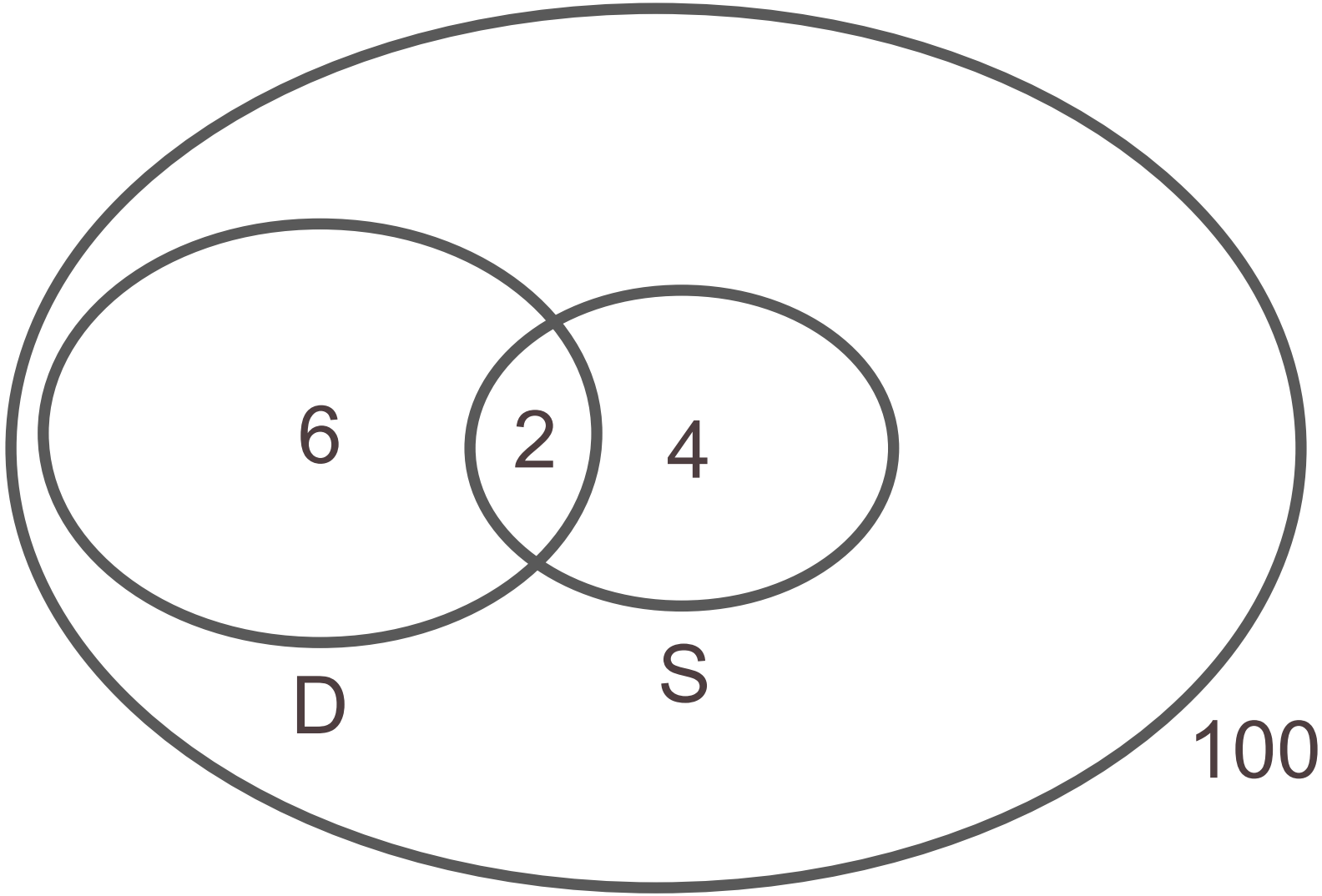
Two parts are selected independently. What is the probability both have D?

4 Conditional

Find $P(D \mid S)$

Given that a part has S, what is the probability it also has D?

Fundamental probability operations



Fundamental probability operations

A compact ladder from basic event algebra to conditioning.

1 Complement

$$P(A^c) = 1 - P(A)$$

Probability that event A does not occur.

2 Sum / OR

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Subtract the intersection because it is counted twice.

3 Product / AND

$$P(A \cap B) = P(A)P(B|A)$$

Joint probability equals a marginal times a conditional.

4 Conditional

$$P(A|B) = P(A \cap B) / P(B)$$

Update the sample space to the cases where B happened.

Two useful special cases

Mutually exclusive events

If $A \cap B = \emptyset$, then
$P(A \cup B) = P(A) + P(B)$

No overlap means the sum rule becomes simple addition.

Independent events

If $P(B|A) = P(B)$, then
$P(A \cap B) = P(A)P(B)$

Knowing A happened does not change the probability of B.

Solution: applying the four rules

Same production-line data: $P(D)=0.06$, $P(S)=0.04$, $P(D \cap S)=0.02$.

1 Complement

$P(D^c)=1-0.06=0.94$

94% of parts have no dimensional defect.

2 Sum / OR

$P(D \cup S)=0.06+0.04-0.02=0.08$

8% have at least one defect. In 1,000 parts, about 80 are affected.

3 Product / AND

$P(D_1 \cap D_2)=0.06 \times 0.06=0.0036$

0.36% of independent pairs contain two parts with D.

4 Conditional

$P(D|S)=0.02/0.04=0.50$

50% of parts with a surface defect also have a dimensional defect.

Interpretation

The overall probability of a dimensional defect is only 6%. But once we know that the part has a surface defect, the probability of also having a dimensional defect rises to 50%.

Why this matters in production

Surface defects may be a warning signal for deeper process problems. Conditional probability quantifies how strongly one defect predicts another.

Manufacturing probability exercise

Question: What is the probability of at least one defective item in two produced items?

Setup: Each item has defect probability $P(D)=1/6$. Assume independent items. **$P(good)=5/6$**

Strategy 1 — Complement

At least one defect is the complement of “no defects”.

P(at least one defect)
$= 1 - P(\text{no defects})$
$= 1 - (5/6 \times 5/6)$
$= 1 - 25/36 = 11/36$

Strategy 2 — Addition rule

Let D_1 = first item defective, D_2 = second item defective.

$P(D_1 \cup D_2)$
$= P(D_1)+P(D_2)-P(D_1 \cap D_2)$
$= 1/6 + 1/6 - (1/6 \times 1/6)$
$= 6/36 + 6/36 - 1/36 = 11/36$

Final answer: $11/36 \approx 30.6\%$

Key idea: “at least one” often becomes easier by using the complement.

From conditional probability to Bayes' rule

Bayes combines the product rule with total probability.

Product rule, written two ways

$$P(A \cap B) = P(A|B)P(B)$$
$$P(A \cap B) = P(B|A)P(A)$$

Same joint event, two factorizations.

Set them equal and solve

$$P(B|A) = P(A|B)P(B) / P(A)$$

This is Bayes' rule in its simplest form.

Expand the denominator

$$P(A) = \Sigma_i P(A|B_i)P(B_i)$$

Law of total probability: average over all possible causes.

Bayes as an update rule

posterior = likelihood × prior / evidence

$$P(B_i|A) = P(A|B_i) \times P(B_i) / \Sigma_j P(A|B_j)P(B_j)$$

$P(B_i\|A)$	=	$P(A\|B_i)$	×	$P(B_i)$	/	$\Sigma_j P(A\|B_j)P(B_j)$
posterior		likelihood		prior		evidence

Interpretation: start with a prior belief, score how compatible the observation is with each cause, then renormalize so all posterior probabilities add to 1.

Bayes' Rule — Manufacturing exercise

Question: a part is found defective. What is the probability it came from Line B?

Setup: Line A makes 70% of parts and has 2% defects. Line B makes 30% of parts and has 8% defects. **A defective part is selected.**

Known probabilities

P(A) = 0.70 **P(B) = 0.30**
P(D | A) = 0.02
P(D | B) = 0.08

We observe: D = "defective part"
We want: P(B | D)

Bayes' rule

$$P(B \mid D) = \frac{P(D \mid B) \cdot P(B)}{P(D \mid A)P(A) + P(D \mid B)P(B)}$$

The denominator is the total probability of observing a defect.

Solution

$P(D) = 0.02 \cdot 0.70 + 0.08 \cdot 0.30$
$= 0.014 + 0.024 = 0.038$

$P(B \mid D) = (0.08 \cdot 0.30) / 0.038$
$= 0.024 / 0.038$
≈ 0.632

Final answer: P(Line B | defective) ≈ 63.2%

Interpretation: even though Line B makes only 30% of parts, it is more likely to be the source of a defective item because its defect rate is higher.

Bayes Rule Exercise: Diagnose the Cause from a Signal

Manufacturing process monitoring: an online sensor raises a warning signal S. What is the most likely cause?

Scenario

A line produces precision parts. A sensor sometimes detects a warning signal:

Signal S = oversized dimension alarm

Possible process states:

T = tool wear
H = temperature drift
M = material variability
N = normal operation

Question: given that signal S appears, which cause is most likely?

Data

Cause / state	Prior P(cause)	Likelihood P(S | cause)
T: tool wear	4%	90%
H: temperature drift	3%	60%
M: material variability	2%	50%
N: normal operation	91%	1%

Use Bayes rule:

$$P(\text{cause} \mid S) = P(S \mid \text{cause})\, P(\text{cause}) / P(S)$$

The signal is the evidence; posteriors tell us the most likely cause.

Solution: From Warning Signal to Most Likely Cause

Bayes rule combines the base rate of each cause with how often that cause produces the signal.

Step 1 — Compute the evidence P(S)

$$P(S) = 0.90 \cdot 0.04 + 0.60 \cdot 0.03 + 0.50 \cdot 0.02 + 0.01 \cdot 0.91$$

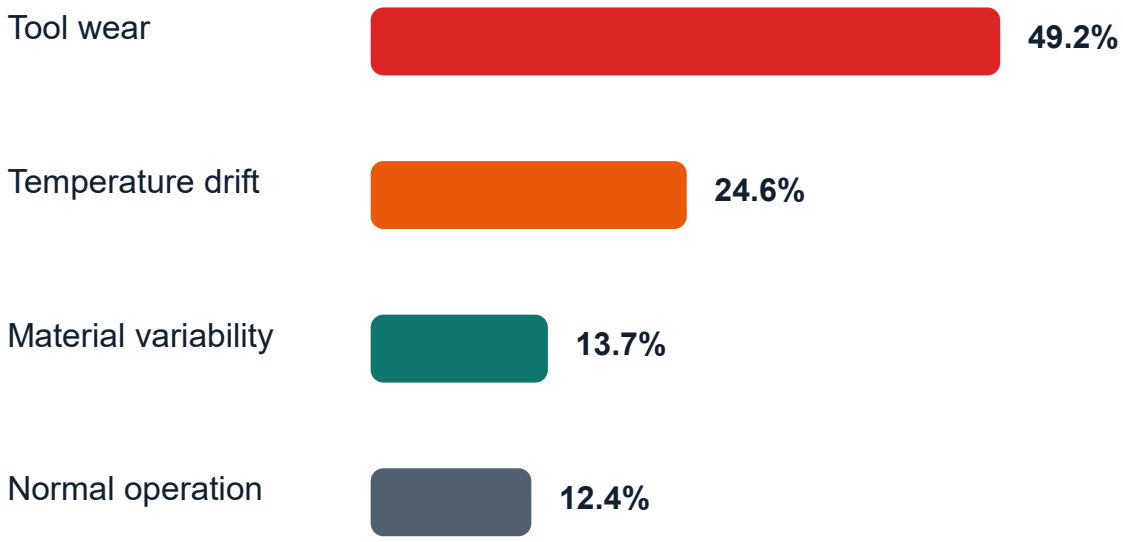
$$P(S) = 0.036 + 0.018 + 0.010 + 0.0091 = 0.0731$$

Step 2 — Compute each posterior

$$P(T \mid S) = (0.90 \cdot 0.04) / 0.0731 = 49.2\%$$
$$P(H \mid S) = (0.60 \cdot 0.03) / 0.0731 = 24.6\%$$
$$P(M \mid S) = (0.50 \cdot 0.02) / 0.0731 = 13.7\%$$
$$P(N \mid S) = (0.01 \cdot 0.91) / 0.0731 = 12.4\%$$

Diagnosis: tool wear is the most likely cause.

Posterior probabilities given signal S



Interpretation: the signal does not just identify a cause; it updates the probability of every possible cause.

Bayes' Theorem

A medical test for a rare disease has a **99%** accuracy (it gives the correct answer in 99% of the cases).
The disease is quite rare, occurring randomly in the general population in only **1 of every 10,000** people.
If **X is positive to the test**, what is the probability that **X actually has the disease**?

Bayes' Theorem

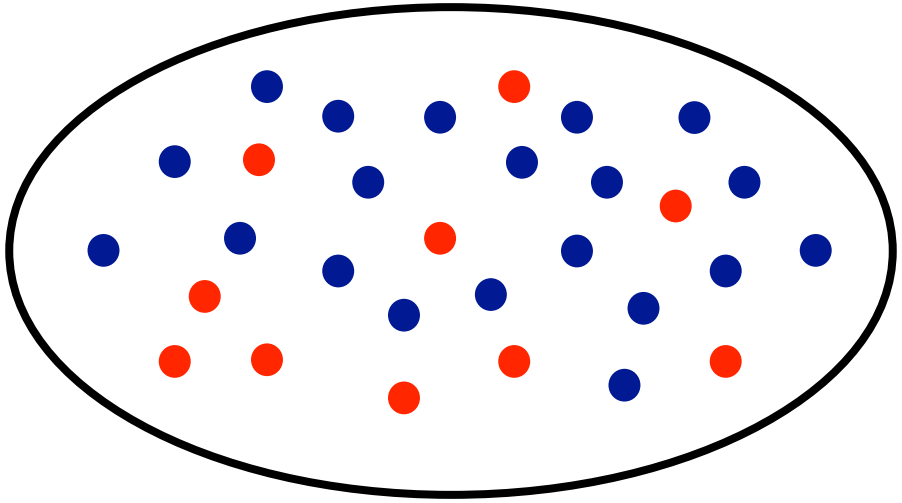
P(Disease|TestPositive) = P(D) P(TP|D) / P(TP)

We know that P(TP|D) = 0.99 and P(D) = 0.0001
P(TP) = P(TP|D)P(D)+P(TP|not D)P(not D)
P(TP) = 0.99 * 0.0001 + 0.01 * 0.9999 = 0.010098

Thus, P(D|TP) = **0.0098!!!**

Sampling in statistical analysis

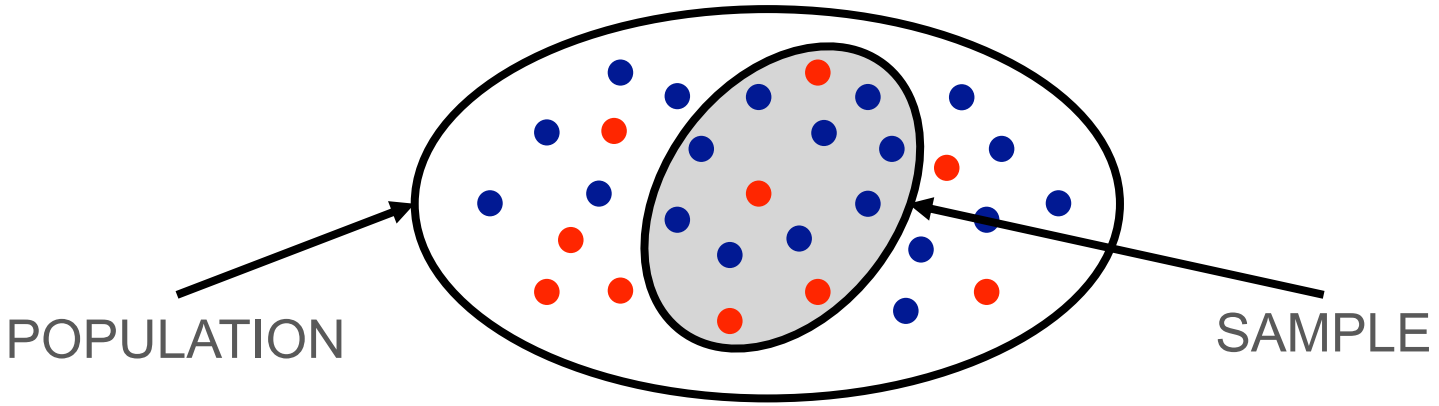
Statistical analysis typically begins with a **set of observations** collected from the phenomenon under study.



They form the foundation for identifying patterns, testing hypotheses, and making predictions.
These observations are often a sample — a subset of the larger population.

Sampling

These observations, named **samples**, most likely represent a **subset** of the whole possible **population** upon which the experiment could be conducted.
The choice of the samples is a **crucial point** for the success and significance of the experiment.



Examples of Samples

Customer Satisfaction Survey

- **Population**: All customers of a company
- **Sample**: 500 randomly selected customers who completed a feedback form

Production Quality Control

- **Population**: All items produced in a factory in one week
- **Sample**: 100 items randomly selected for inspection

Employee Productivity Study

- **Population**: All employees in a multinational company
- **Sample**: 50 employees from three departments

Examples of Samples

Market Research

- **Population**: All potential buyers of a new product
- **Sample**: 1,000 respondents from an online survey

Website Analytics

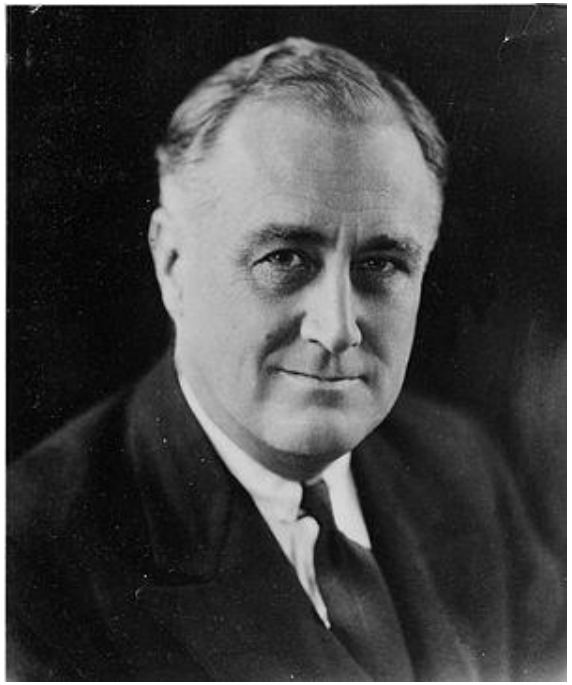
- **Population**: All visitors to a website over a month
- **Sample**: Data from 10,000 randomly selected sessions

Energy Consumption Analysis

- **Population**: All households in a city
- **Sample**: 200 households with smart meters

Sampling

1936 U.S. Presidential Elections



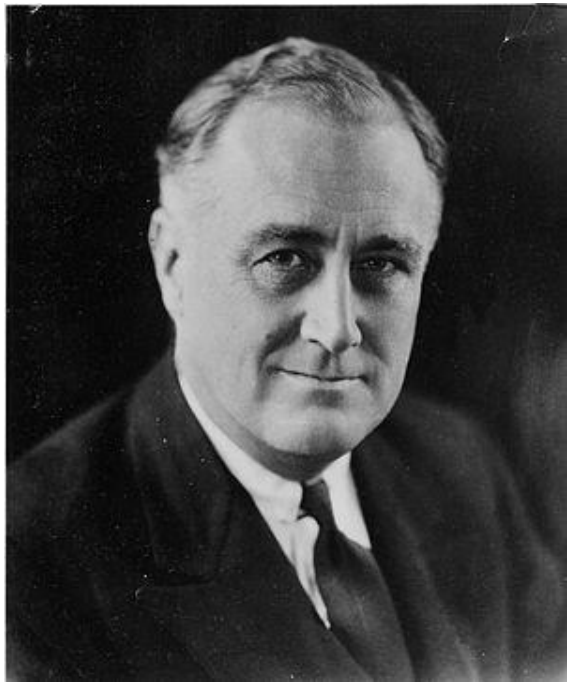
Franklin Delano Roosevelt
(DEMOCRAT)



Alfred Landon
(REPUBLICAN)

Sampling

1936 U.S. Presidential Elections

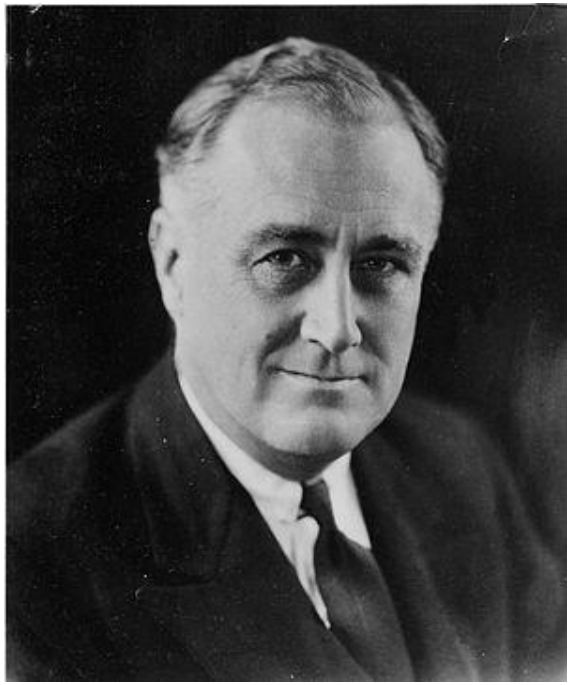




The magazine *The Literary Digest* conducted a poll that since 1920 had always correctly predicted the outcome of the elections

Sampling

1936 U.S. Presidential Elections

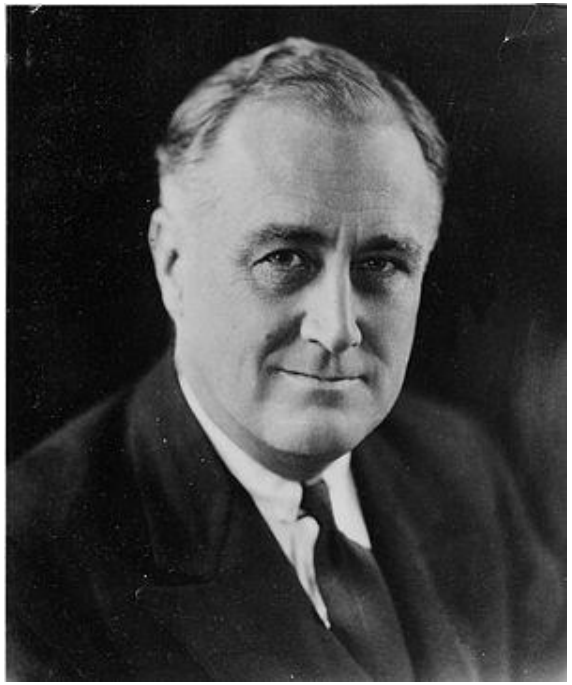




In 1936 it predicted **Landon 370** vs. **Roosvelt 161**

Sampling

1936 U.S. Presidential Elections





The real outcome: **Landon 8** vs. **Roosvelt 523!**

What Went Wrong?

What caused such a huge mistake?

The **sample (10M) of interviewers** were taken from:

- Telephone directories
- Club membership lists
- Magazine subscribers
- Car owners

Sampling

What caused such a huge mistake?

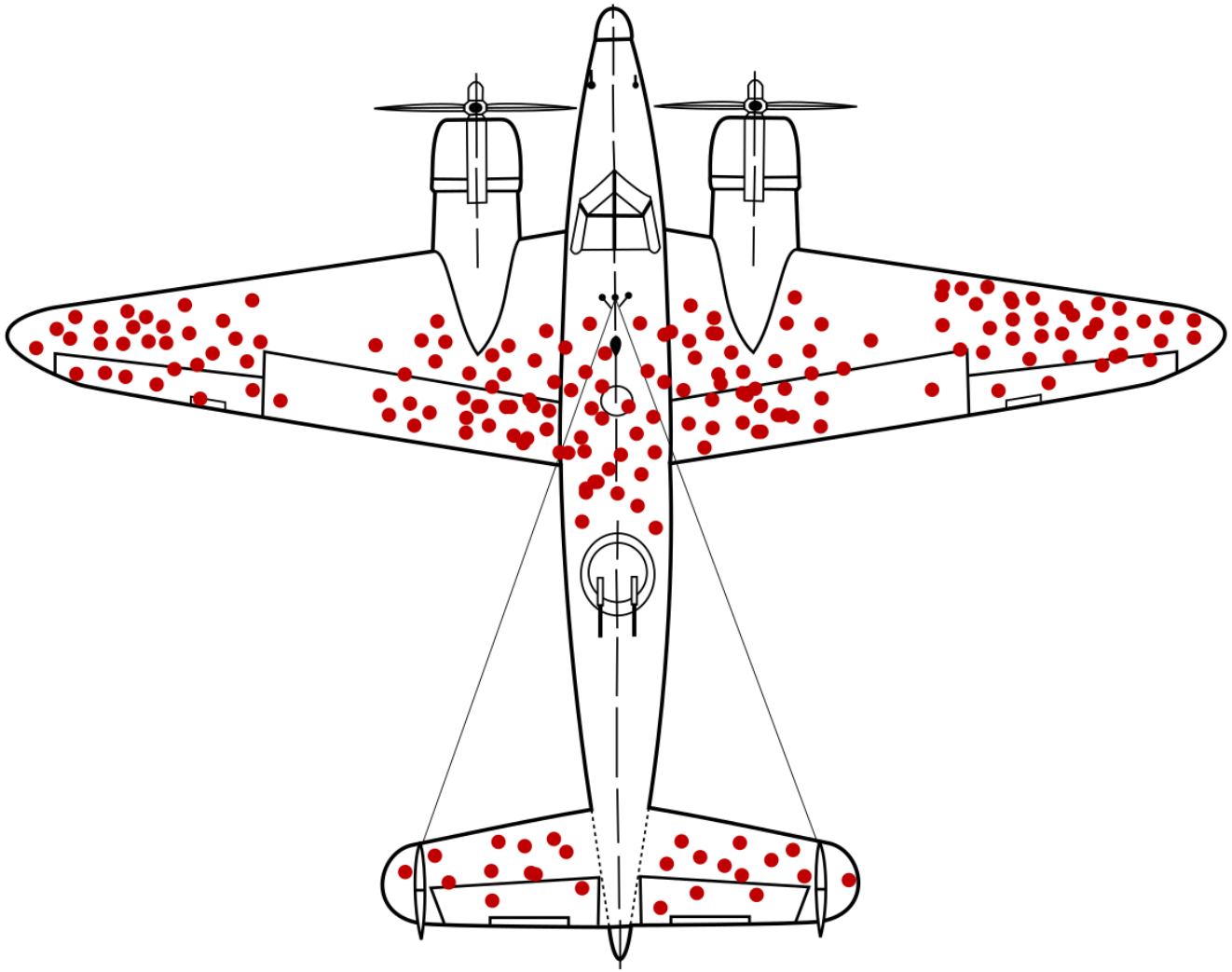
Sampling bias

- Post 1929 crisis, telephones/cars were for the rich
- Interviewers coming from upper-/middle-class

Non-response bias

- Only 2.4M answers out of 10M interviewers

Sampling Bias



Lesson learned

A **large sample** is not enough — it must be **representative** of the population.

Sample Statistic

A **sample statistic** is any **numerical function** computed from a **random sample**.
It is used to **summarize**, **estimate**, or **test** properties of the population.

Examples:

- Mean (Expected Value)
- Median
- Variance
- Standard deviation
- Correlation coefficient

Sample Statistics

Mean of a set of equally weighted samples

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \mathrm{E}[X] = \sum_i x_i \cdot p(x_i).$$

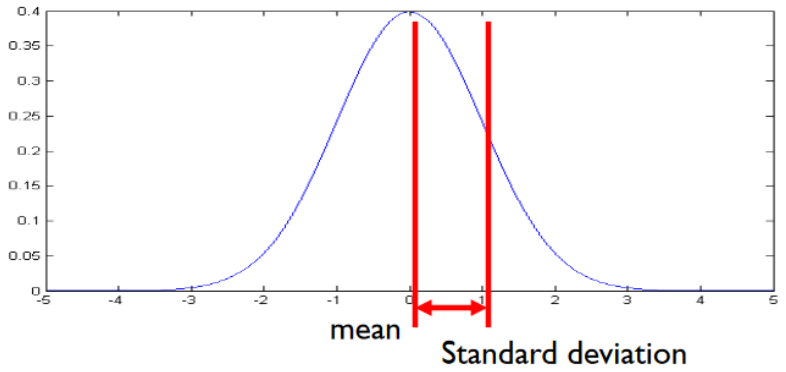
Median The median is the middle point in a dataset—half of the data points are smaller than the median and half of the data points are larger.

Variance of a set of equally weighted samples

$$\sigma_x^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad \operatorname{Var}(X) = \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] \qquad \operatorname{Var}(X) = \sum_{i=1}^{n} p_i \cdot (x_i - \bar{x})^2$$

Standard deviation of a set of equally weighted samples

$$\sigma_x = \sqrt{\sigma_x^2}$$



Example

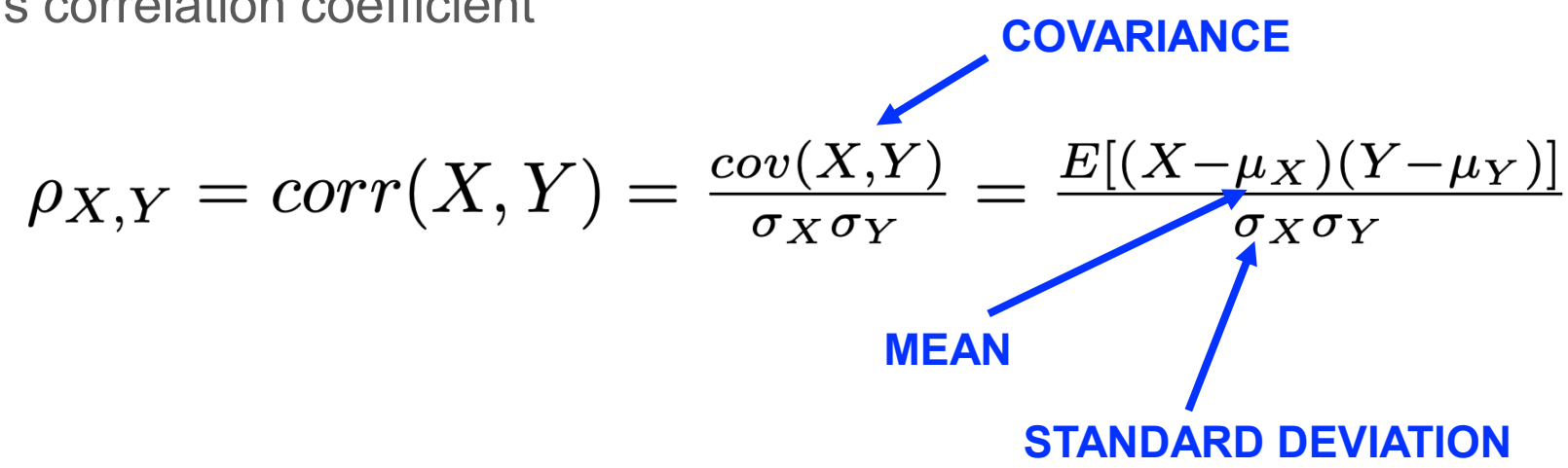
DATA DI INIZIO MISSIONE	DATA DI RIENTRO	DATA CONSEGNA DOCUMENTAZIONE IN SEGRETERIA	DATA DI PRESA IN CARICO=CONTROLLO DOCUMENTAZIONE (EVENTUALE RICHIESTA INTEGRAZIONE)	DATA DI COMPLETAMENTO PRATICA (COMPLETA E VA IN CODA DA LIQUIDARE SE CON INTEGRAZIONI)	DATA EMISSIONE ORDINATIVO DI PAGAMENTO
02/07/25	06/07/25	24/07/25	02/09/25	02/09/25	02/09/25
27/07/25	31/07/25	07/08/25	07/08/25	04/09/25	03/10/25
24/06/25	24/06/25	04/09/25	04/09/25	09/09/25	12/09/25
21/06/25	28/06/25	04/09/25	04/09/25	17/09/25	18/09/25
26/07/25	03/08/25	04/09/25	04/09/25	16/09/25	18/09/25
26/07/25	02/08/25	04/09/25	04/09/25	09/09/25	12/09/25
25/05/25	29/05/25	04/09/25	04/09/25	21/10/25	23/10/25
13/07/25	18/07/25	24/07/25	04/09/25	04/09/25	04/09/25
15/07/25	20/07/25	27/07/25	05/09/25	05/09/25	05/09/25
03/07/25	04/07/25	29/07/25	05/09/25	05/09/25	05/09/25

- How many days between the return from a mission and reimbursement?
- Compute **mean** and **median** and distribution
- Discuss

Correlation

Measuring correlation between two random variables

Pearson's correlation coefficient



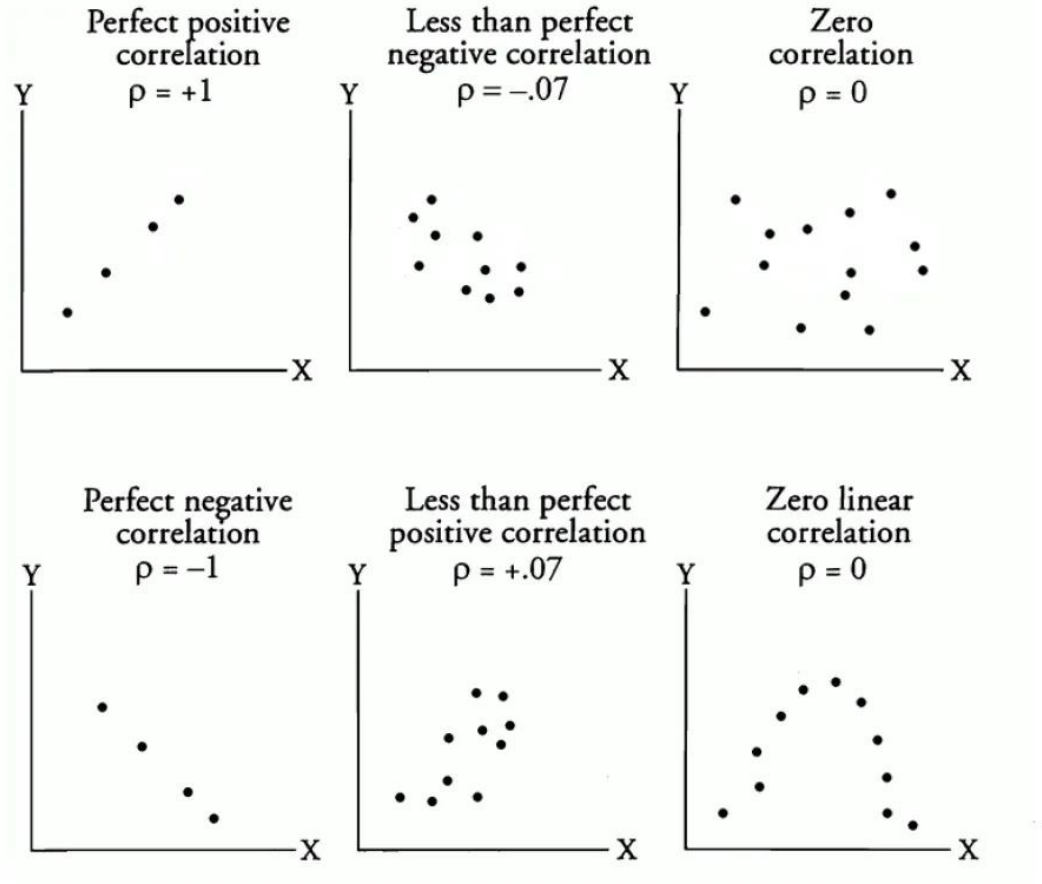
Correlation measures the strength of the linear relationship between two random variables

Correlation

The correlation coefficient lies in the [-1,1] interval

- A value of 1 implies a **positive linear** correlation
- A value of -1 implies a **negative linear** correlation
- A value of 0 implies **no correlation**

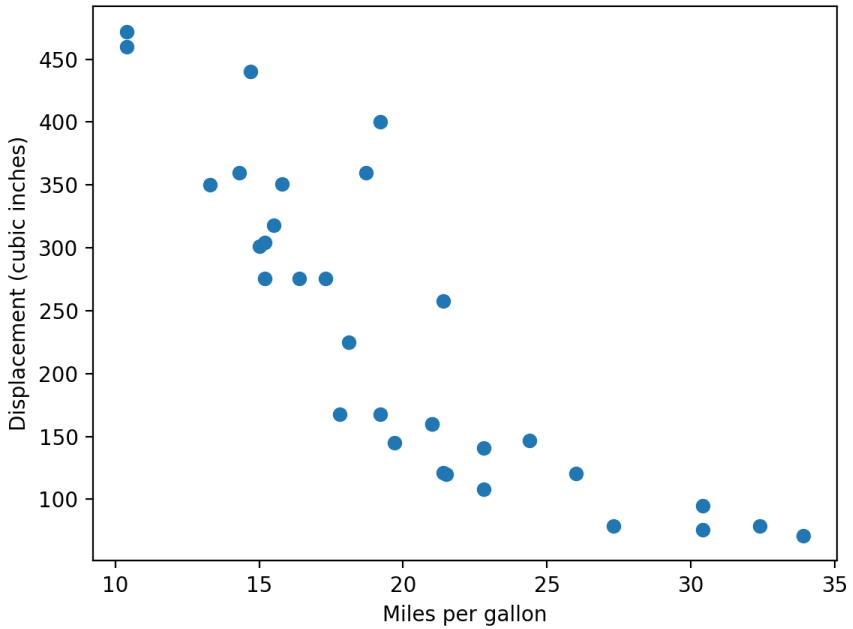
Correlation



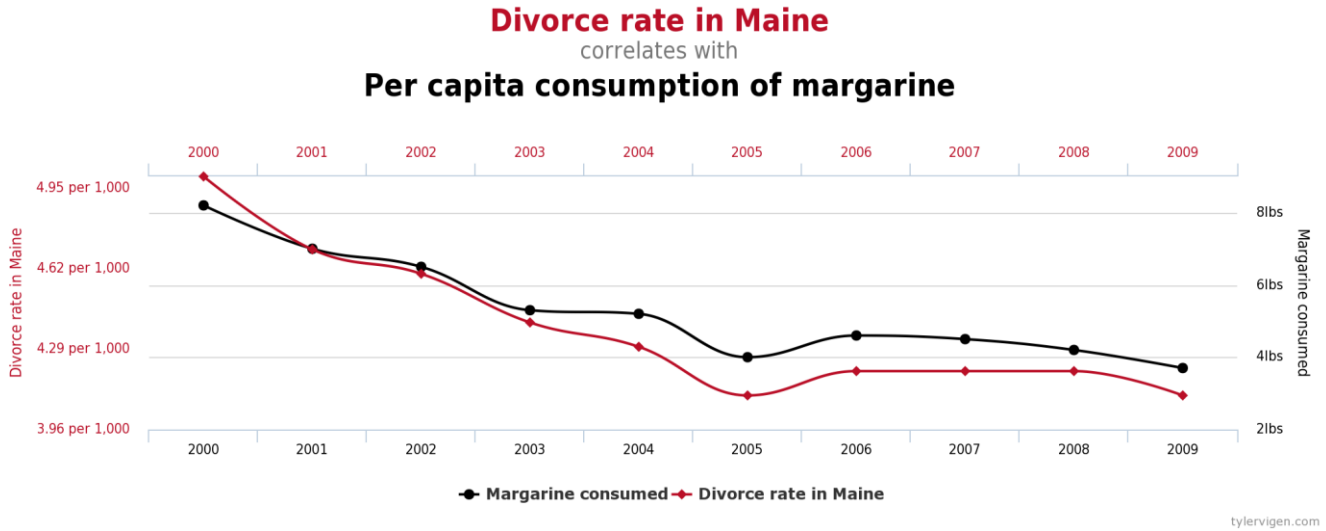
Correlation

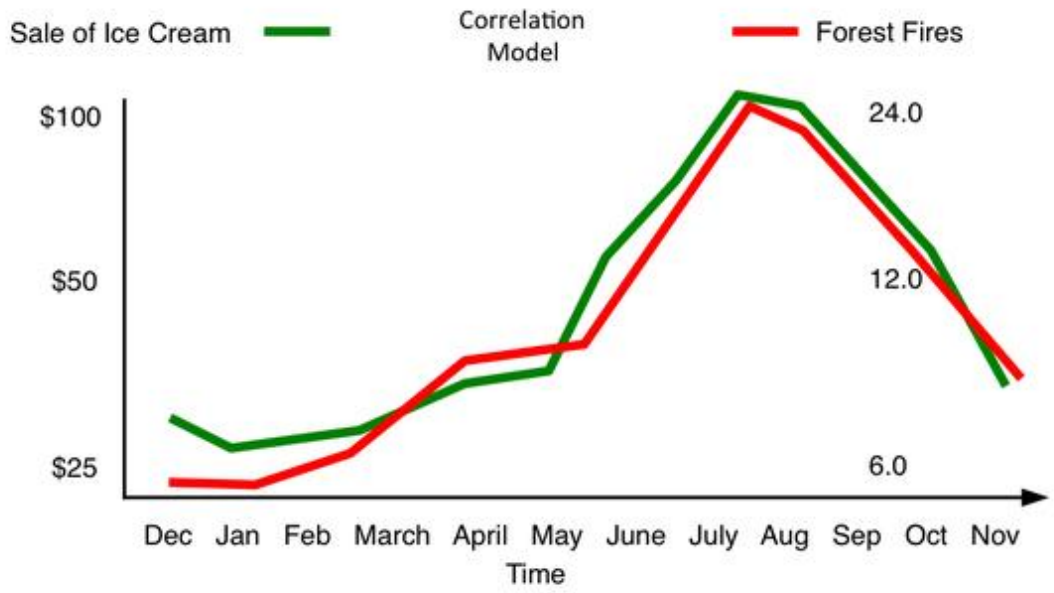
Example

- `mtcars` dataset (Motor Trend 1974)
- X = mpg (miles per gallon)
- Y = disp (displacement)
- cor(X,Y) = -0.8475514



Correlation ≠ Causation





Correlation ≠ Causation

A common mistake in data analysis is to confuse correlation with causation.

Just because B happens after A, it doesn't mean that A causes B.

Possible Scenarios

- A causes B
  - e.g., Increasing production speed leads to higher defect rates.
- B causes A
  - e.g., Higher customer complaints lead to more quality checks.
- Coincidence
  - e.g., Ice cream sales and forest fires both increase in summer.
- Third variable C causes both A and B
  - e.g., Economic growth (C) increases both advertising (A) and sales (B).

Spurious Correlation: When Data Misleads

Spurious correlation occurs when two variables appear to be related, but the connection is purely coincidental or driven by a hidden third factor.

Why It Matters in Data Science and Engineering
Misinterpreting data can lead to **poor strategic decisions**, **wasted resources**, or **false assumptions**.

Always ask:
Is there a logical mechanism behind the correlation?
Could a third variable be influencing both?

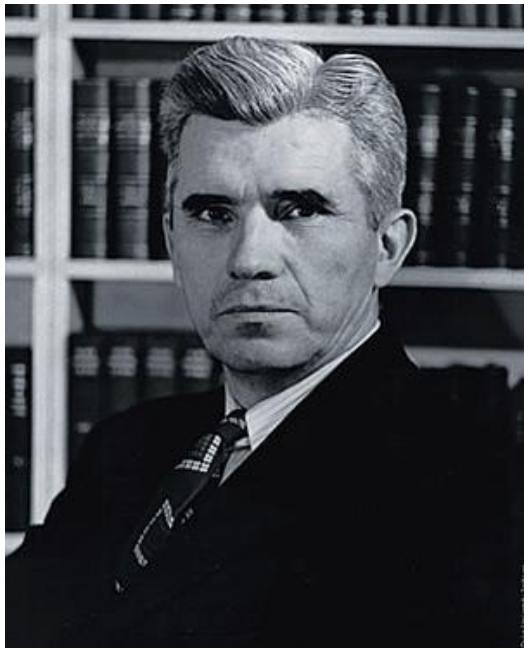
Bonferroni’s Principle: Beware of Meaningless Patterns

In **big data analysis**, the more patterns you test, the more likely you are to find **something that looks significant** — even if it’s **not**.

In Data Science and Engineering, testing many hypotheses (e.g., customer behaviors, market trends) can lead to false discoveries.

Bonferroni’s correction adjusts for this by lowering the significance threshold when multiple tests are performed

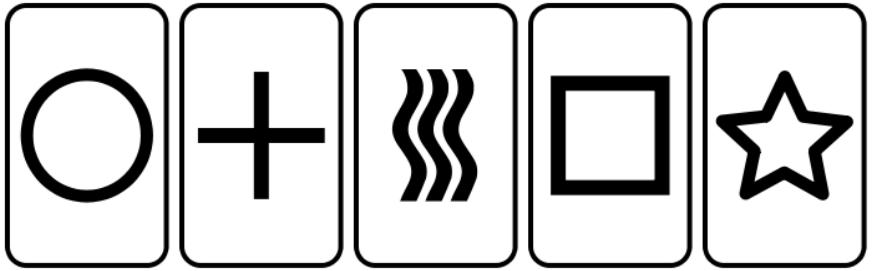
Joseph Rhine & the ESP Hypothesis



[Example from Luskovec et al. “Mining Massive DataBases”]

In the 1950s, **Joseph Rhine**, a parapsychologist, hypothesized that some individuals possessed **Extra-Sensory Perception (ESP)** — the ability to perceive information beyond the known senses.

His Experiment



Subjects were asked to guess hidden symbols on Zener cards.
Out of thousands of participants, a few guessed long sequences correctly.
Rhine believed these individuals had ESP.

What Went Wrong?



- In follow-up tests, their performance dropped below chance.
- Rhine concluded that **telling people they had ESP made them lose it**.

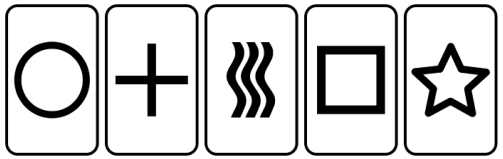
Statistical Insight: Bonferroni's Principle

When testing **many hypotheses**, some will appear significant **just by chance**.
Rhine's results were likely **false positives** due to **multiple testing**.
This illustrates the need for **correcting significance thresholds** when analyzing large datasets.

Probability exercise: guessing cards

Question: With 5 cards sampled with replacement, what is the probability of guessing all N cards after repeating the experiment M times?

Setup: Each card has 5 possible values. Sampling is with replacement, so guesses are independent.
Single experiment: guess all N cards correctly ⇒ **p = (1/5)^N**



Each guess has probability 1/5 of being correct.

Strategy 1 — Complement

Probability of succeeding at least once in M independent repetitions:

P(at least one success)
= 1 − P(no success in M trials)
= 1 − (1 − p)^M
= 1 − [1 − (1/5)^N]^M

N=8, M=2000 → P = 50%

Interpretation of results

A crucial issue when dealing with statistical analysis is clearly to **provide an interpretation of the results**

Examples:

- Is a given conjecture/hypothesis **verified** by data?
- Is method A to be **preferred** over B on task T?

Simpson’s paradox

We want to assess whether **treatment A** is better* than **treatment B** as a cure for kidney stones

*We will see later in the course what **better** means…

Simpson’s paradox

Overall, **Treatment B** has a higher success rate than Treatment A



NUMBER OF HEALED CASES	Treatment A	Treatment B
Total	273/350 (78.0%)	289/350 (82.6%)

Simpson’s paradox

For both **small stones** and **large stones**, **Treatment A** has a higher success rate than Treatment B

NUMBER OF HEALED CASES	Treatment A	Treatment B
Small stones	81/87 (93.1%)	234/270 (86.7%)
Large stones	192/263 (73.0%)	55/80 (68.8%)
Total	**273/350 (78.0%)**	**289/350 (82.6%)**

Simpson’s paradox

Which is the best treatment for the disease?

- A is best according to the **partial** measurement
- B is best according to the **global** measurements

We speak of Simpson’s paradox in statistics when a trend can be **observed** in some groups of data, but it **disappears** when such groups are **combined**.

Confounding Variable

A variable that **influences both** the independent variable (e.g., treatment choice) and the dependent variable (e.g., recovery rate), potentially distorting the apparent relationship between them.

In the kidney stone study:

- **Independent variable**: Treatment (A or B)
- **Dependent variable**: Recovery (healed or not)
- **Confounding variable**: Stone size (small or large)

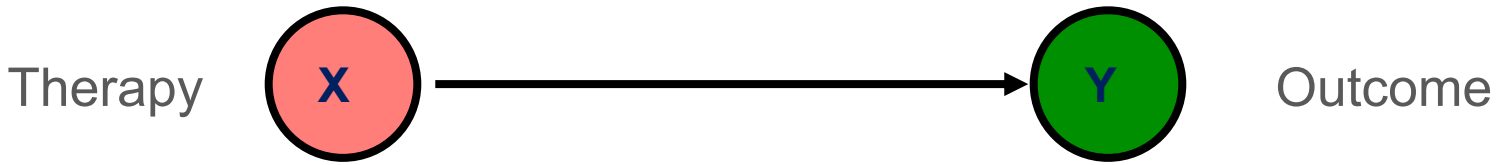
Confounding Variable

Stone size affects:

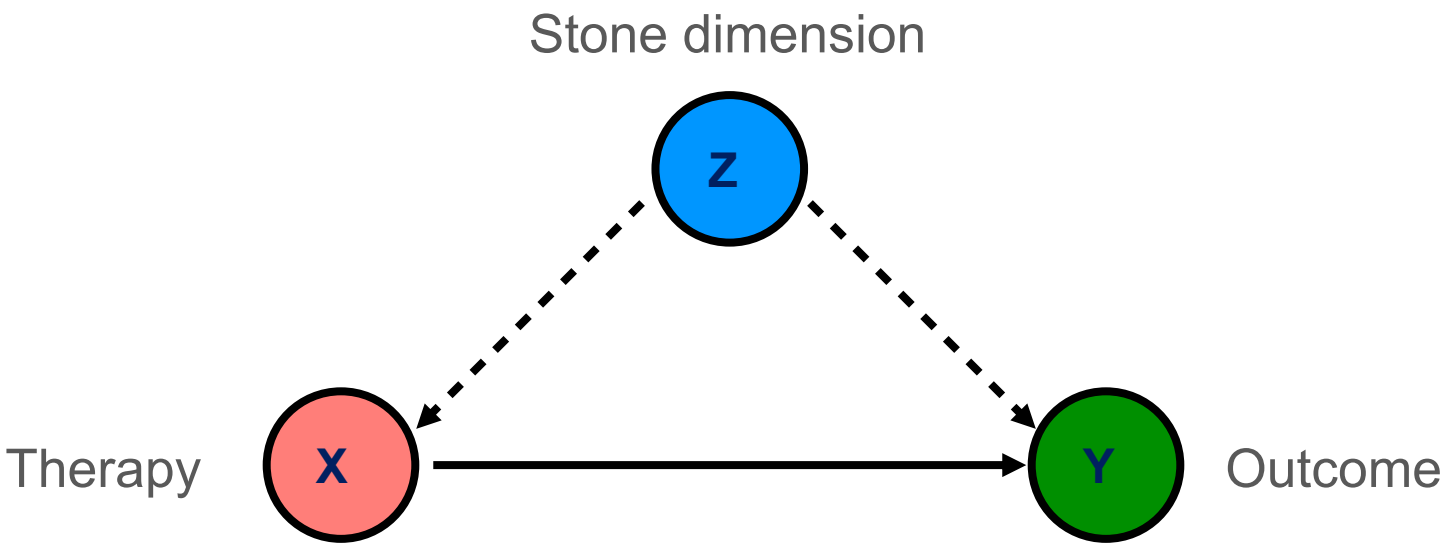
- The **choice of treatment** (e.g., simpler cases more often treated with B).
- The **likelihood of recovery** (smaller stones are easier to treat).

This leads to Simpson's Paradox: although Treatment A is better in both subgroups, Treatment B appears better overall due to the **uneven distribution of stone sizes across treatments**.

Simpson’s paradox

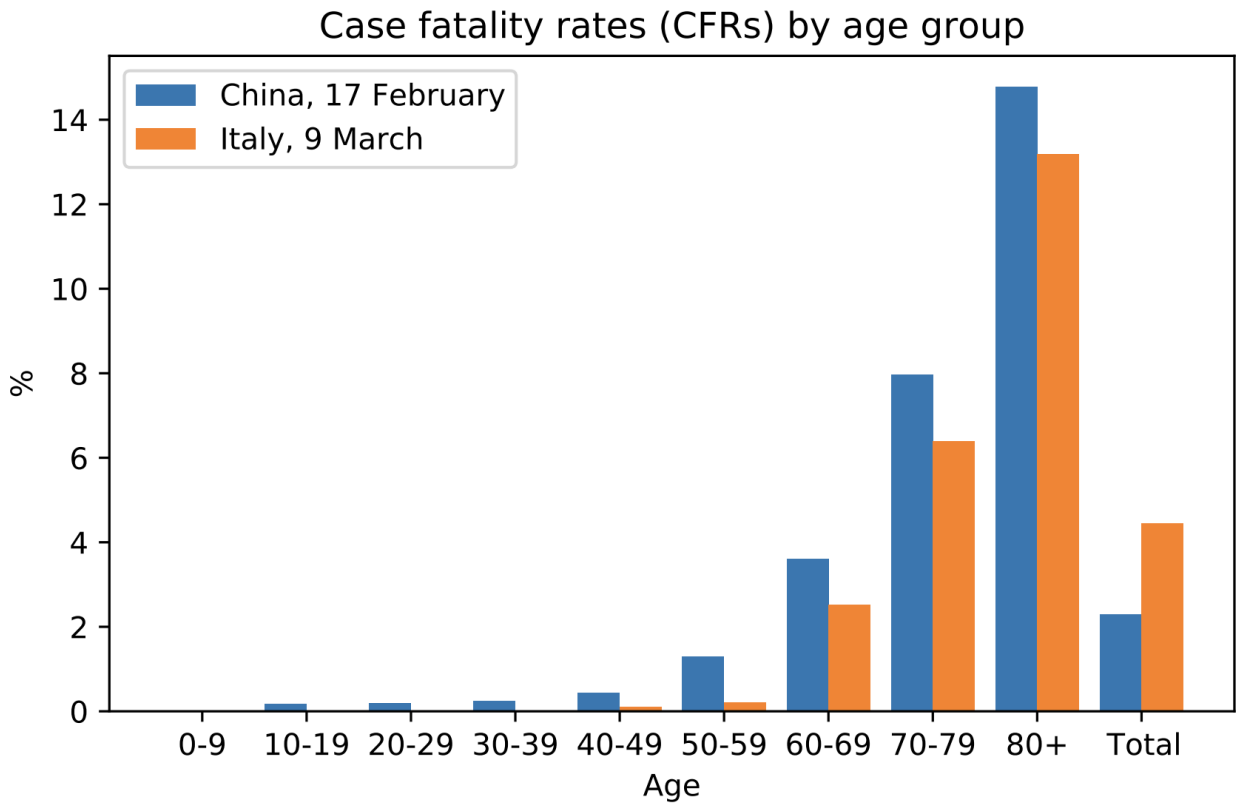


Simpson’s paradox



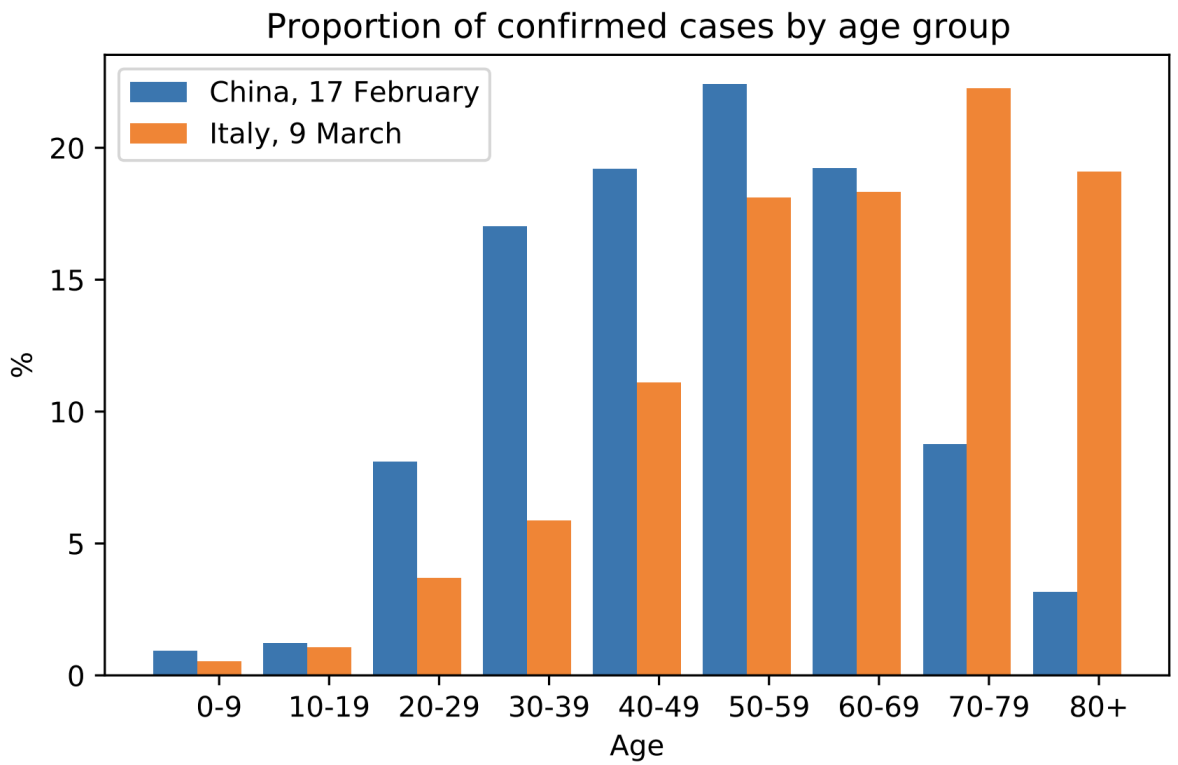
Simpson’s paradox

An example with COVID-19 cases

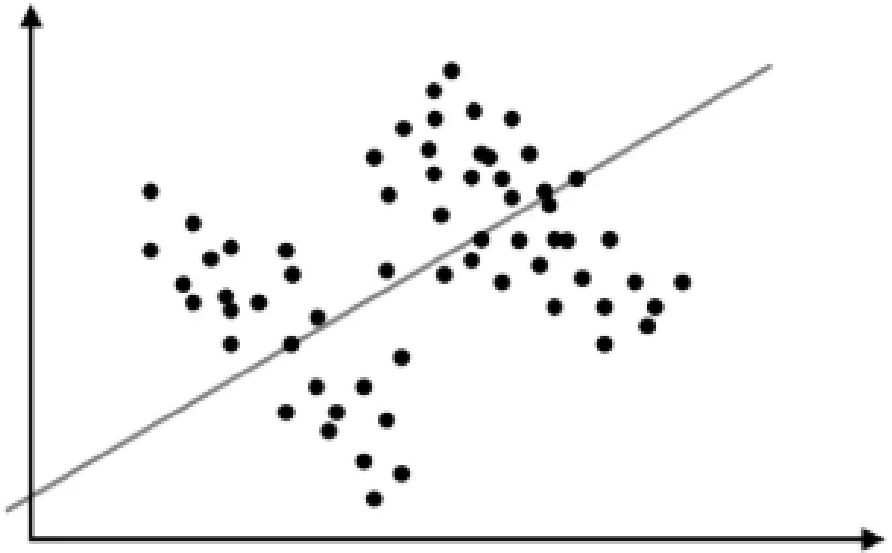


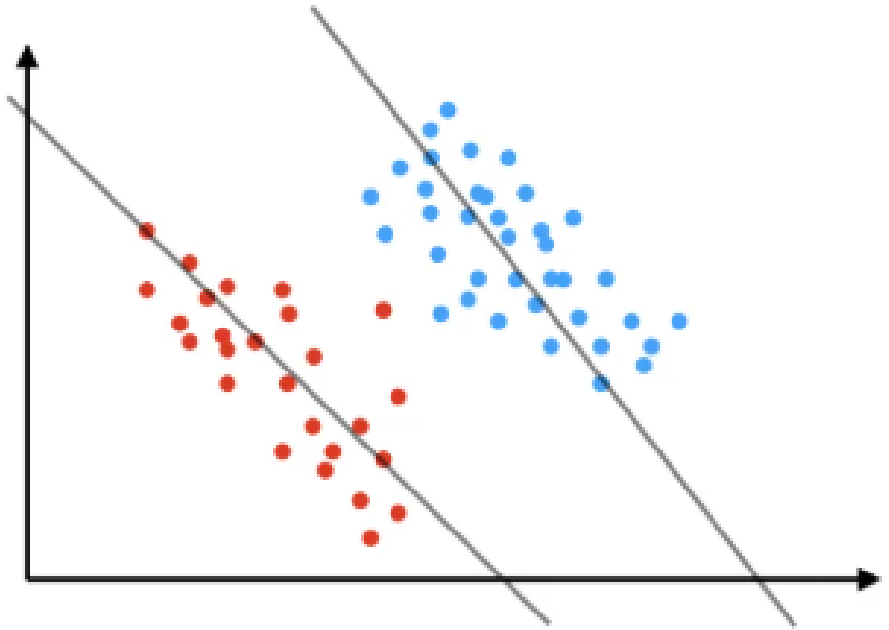
Simpson’s paradox

An example with COVID-19 cases



Simpson’s Paradox





Simpson’s paradox

Implications on decision making

- Should we **always** prefer **partitioned** results?

Not necessarily!

- If so, why not **arbitrarily** partition data?
- Take into account **meaningful** external variables!

Sure-Thing Principle

"An action A that **increases** the probability of an event E in each **subpopulation** must also increase the probability of E in the **population as a whole**, **provided only that** the action does **not** change the distribution of the subpopulations."

[J. Pearl 2009]

Descriptive vs. Inferential

Descriptive statistics

- It has the goal of providing a **summary** of the observed phenomena (with no further assumptions)
- No aim to make conclusions **beyond** the given data
- Useful to **quickly** analyze/visualize data

Examples: mean, minimum, maximum, median, mode, standard deviation, kurtosis, skewness

Inferential statistics

- It has the goal of **generalizing** results provided from given data to the **whole existing population**
- Rigorous **hypothesis testing**
- Build a **model** of data and **draw** conclusions from it

Inferential Statistics

Example: collect height of all students in this class

Descriptive statistics

- I can say that the mean value is 170 cm.

Inferential statistics

- Can I say that the mean value of Italian young people between 20 and 30 years is 170 cm up to some level of confidence?

Inferential Statistics

Frequentist approaches

- Formulate hypothesis H
- Estimate a confidence level T
- Repeat experiment N times to gather data
- Compute a test statistics S with the N samples
- **Accept or reject hypothesis H according to S and T**

Bayesian approaches

- Apply probability to statistical problems
- **Find out the probability of a hypothesis H**
- Exploit Bayes' theorem
- Based on the concept of prior probability

Hypothesis Testing

A widely used **frequentist** paradigm

Data science very often has to deal with

- Formulate a hypothesis/theory/conjecture
- Collect data to assess the hypothesis
- Process data (cleaning, aggregation, visualization)
- **Test the significance of the hypothesis**

The goal is to evaluate whether a certain hypothesis is **“likely to be true”** given the data we have

Hypothesis Testing

Examples

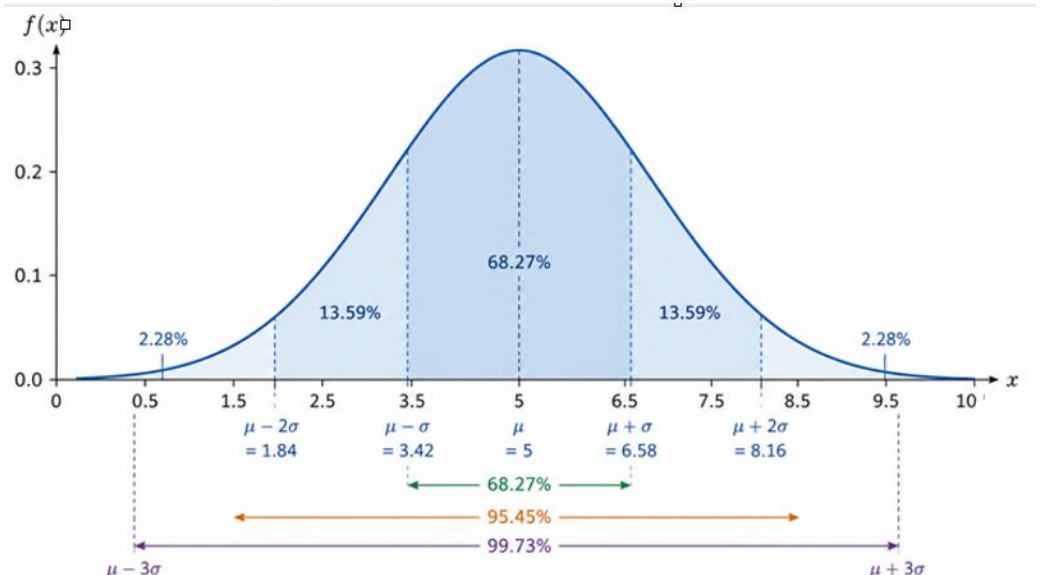
- Is traffic flow in a road network at a given time dependent on whether a day is a weekday/holiday?
- Suppose we build two machine learning classifiers A and B to predict traffic flow. How can we assess whether A is better than B?
- We hypothesize that the average weight of 1-year children is 9.1 kg. From a sample of 100 children we get a mean of 9.2 kg. What can we conclude?

Hypothesis Testing – The Idea

- I have a coin. My **hypothesis** is that it is a fair coin, but I what to **test** it!

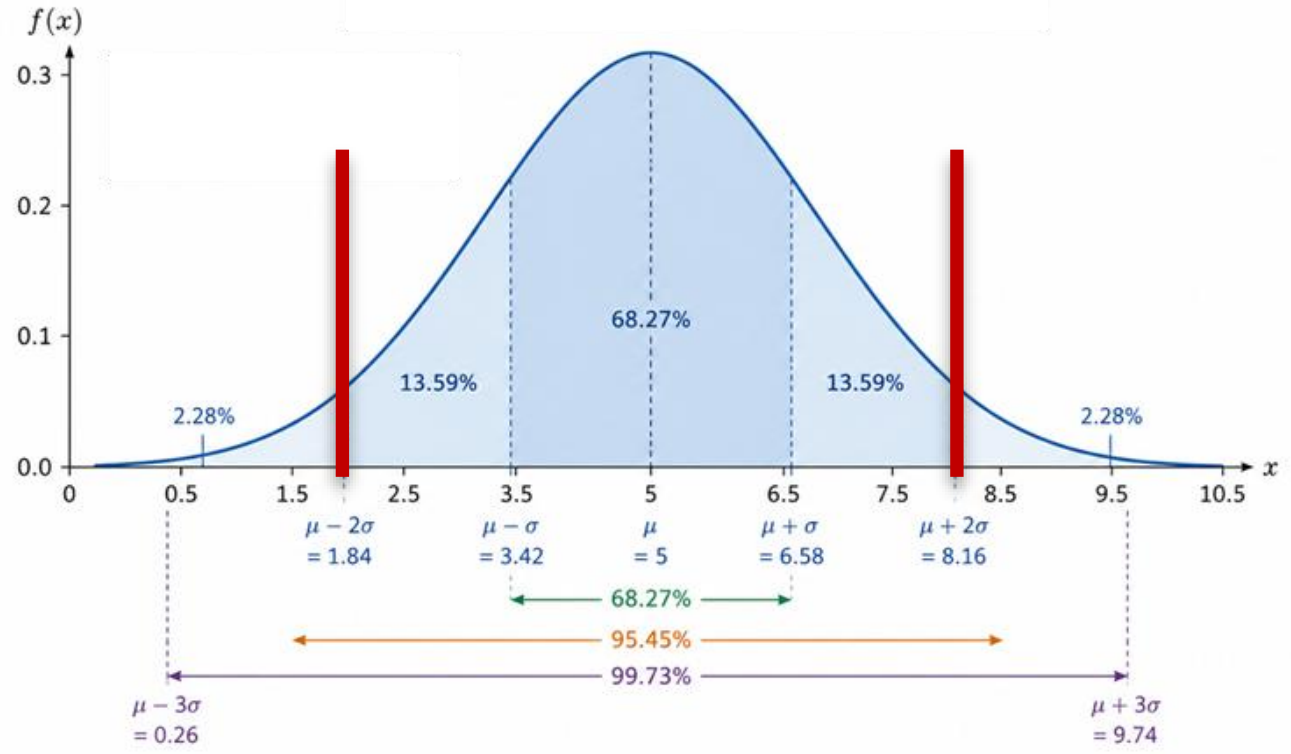
- Test Idea:
  - Launch the coin 10 times and count the number of head
  - Repeat 10 times and average the count

- What is the probability distribution of the outcomes of this experiment? **This is a theoretical probability, we compute it without data. We call it test distribution / test statistics. It depends on the test, the number of repetitions, etc.**



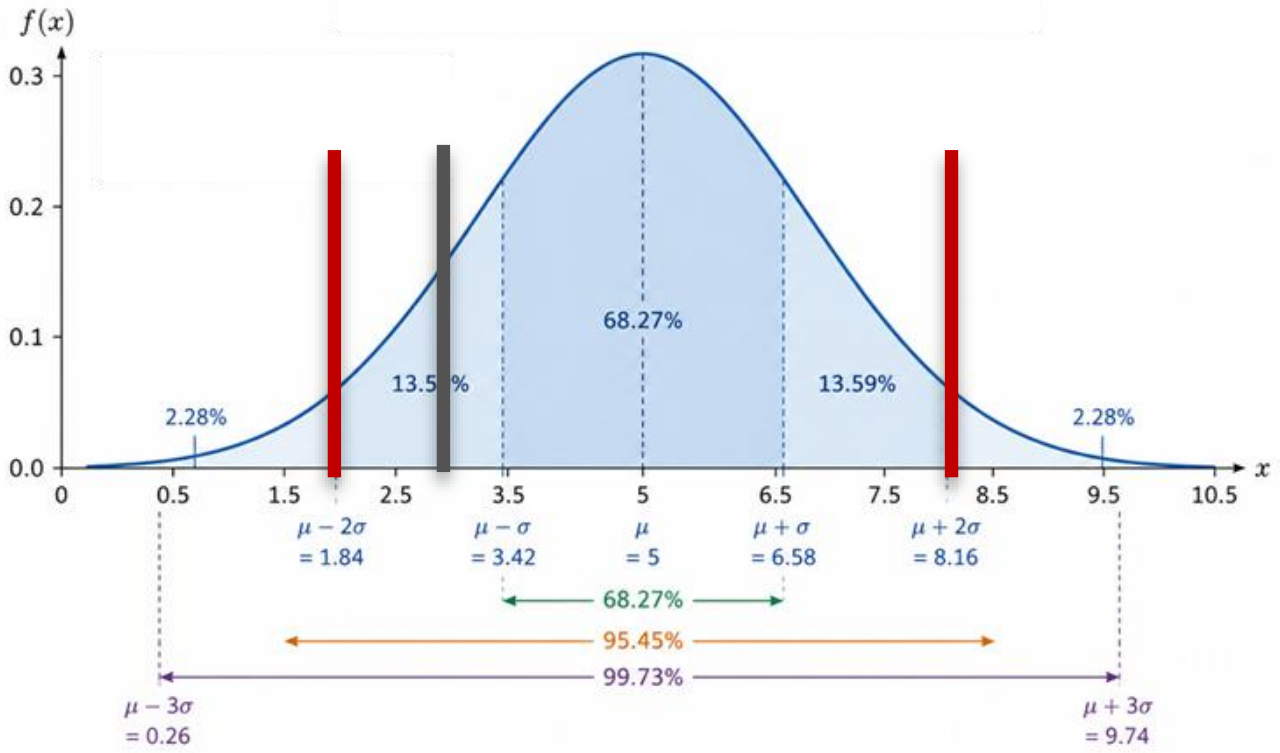
Hypothesis Testing – The Idea

- Now, I select a threshold, If I get on average more than 8 or less than 2 heads, it is too suspicious…



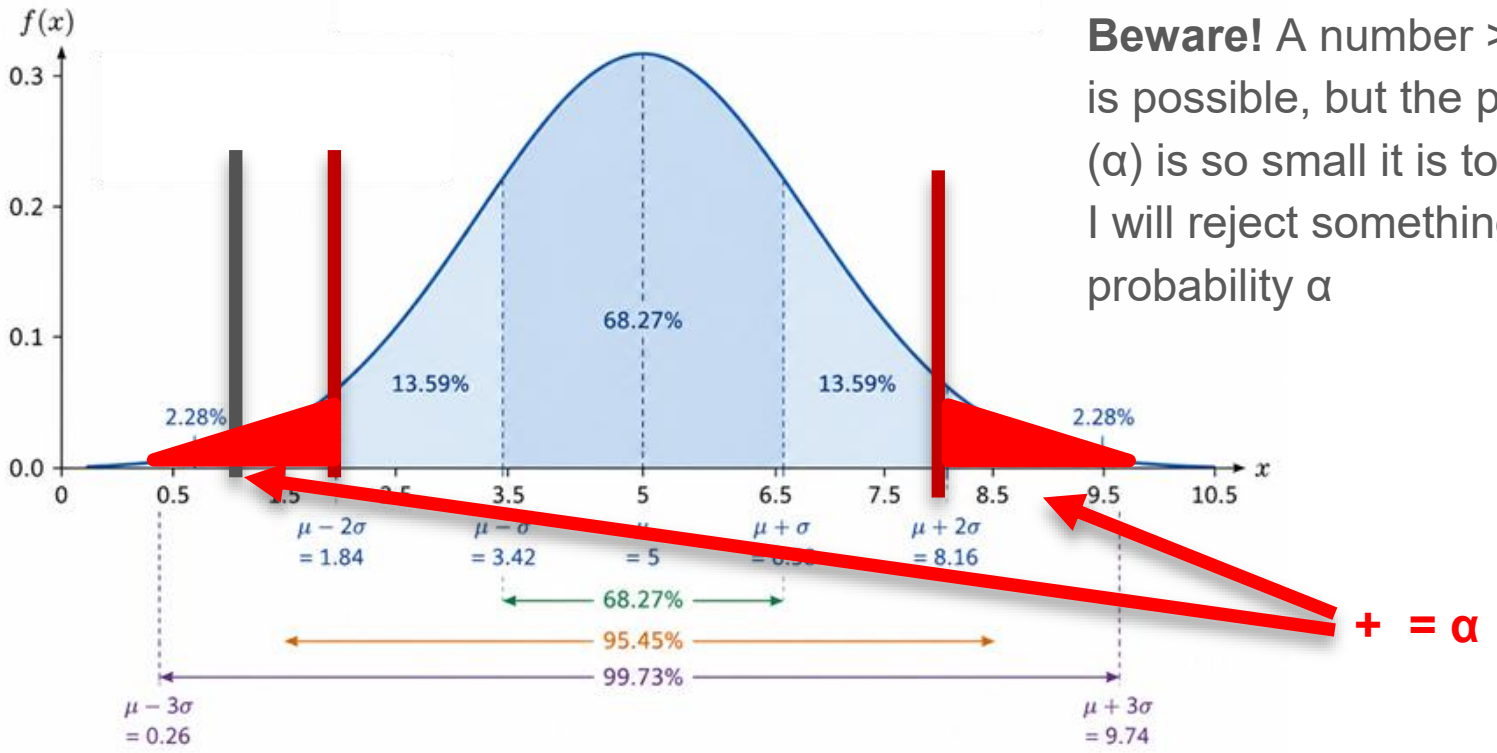
Hypothesis Testing – The Idea

- Then I run the experiment. I get 3!
- I am a bit unlucky, but based on the threshold I set, I cannot rule out (reject) the hypothesis that the coin is fair



Hypothesis Testing – The Idea

- Then I run the experiment. I get 1!
- It is so strange that based on the threshold I set, I rule out (reject) the hypothesis that the coin is fair



Beware! A number > 8 or < 2 is possible, but the probability (α) is so small it is too strange. I will reject something legit with probability α

Hypothesis Testing

Which is the **appropriate** statistical test?

It depends on…

- Which is the **task**
- Which is the **data distribution**
- How **large** is the **sample**

Hypothesis Testing

We first formulate a hypothesis about data, which we name the **null hypothesis H_0**.

The **alternative hypothesis H_1** is typically just the opposite of **H_0**, i.e. "H_0 is not true".

	H_0 true	H_1 true
Reject H_1	**OK**	**Type II Error (False Negative)**
Reject H_0	**Type I Error (False Positive)**	**OK**

A Type I error is the decision to reject the hypothesis H_0 when it is true
A Type II error is the decision to accept the hypothesis H_0 when it is false

Hypothesis Testing

The **power of the test (1-β)** is the probability of rejecting H_0 in the case that H_1 is true (the higher, the better)

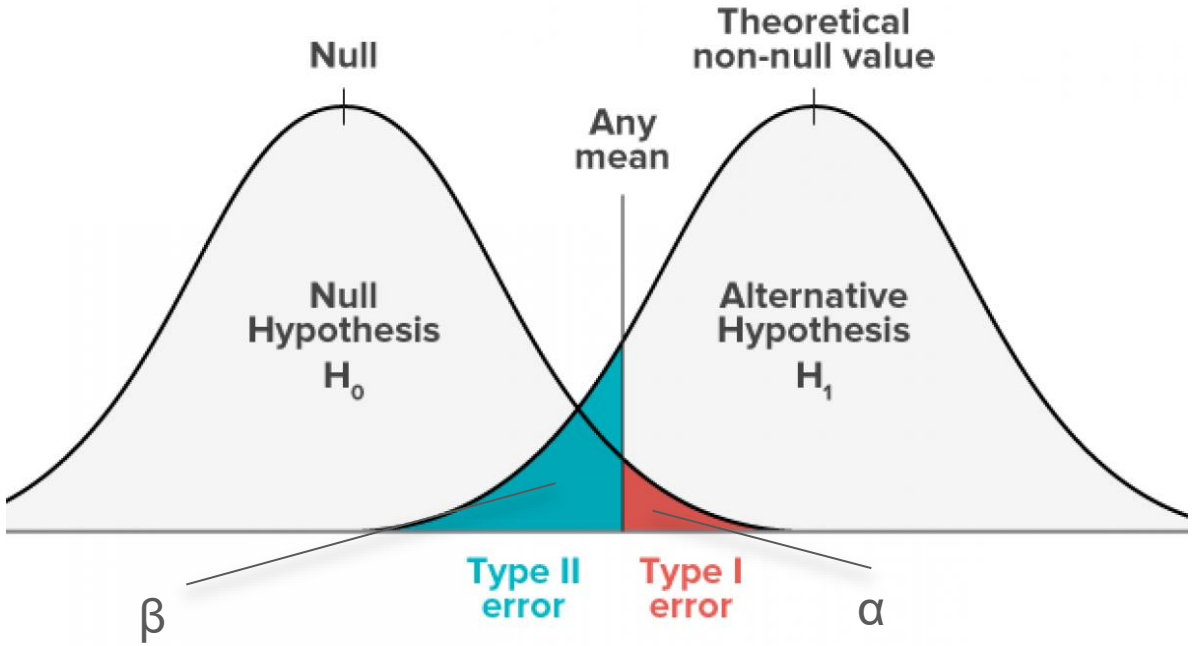
The **size of the test (α)** is the probability of rejecting H_0 in the case that H_0 is true (the lower, the better)

	H_0 true	H_1 true
Reject H_1	$1-\alpha$	β
Reject H_0	α	$1-\beta$

POWER

SIZE

Hypothesis Testing



- Alpha (α) is the level of significance, or the probability of making a Type I error. A Type I error occurs when we reject the null hypothesis when it is actually true.
- Beta (β) is the probability of making a Type II error, which occurs when we fail to reject the null hypothesis when it is actually false. In other words, we are saying that there is no difference in the population parameter when there is.

P-Value

The classic approach to hypothesis testing is to compute the **probability of observing "a result at least as extreme as the one observed", given that the null hypothesis is true**

$$p = \Pr(\text{a result at least as extreme as the one observed} \mid H_0).$$

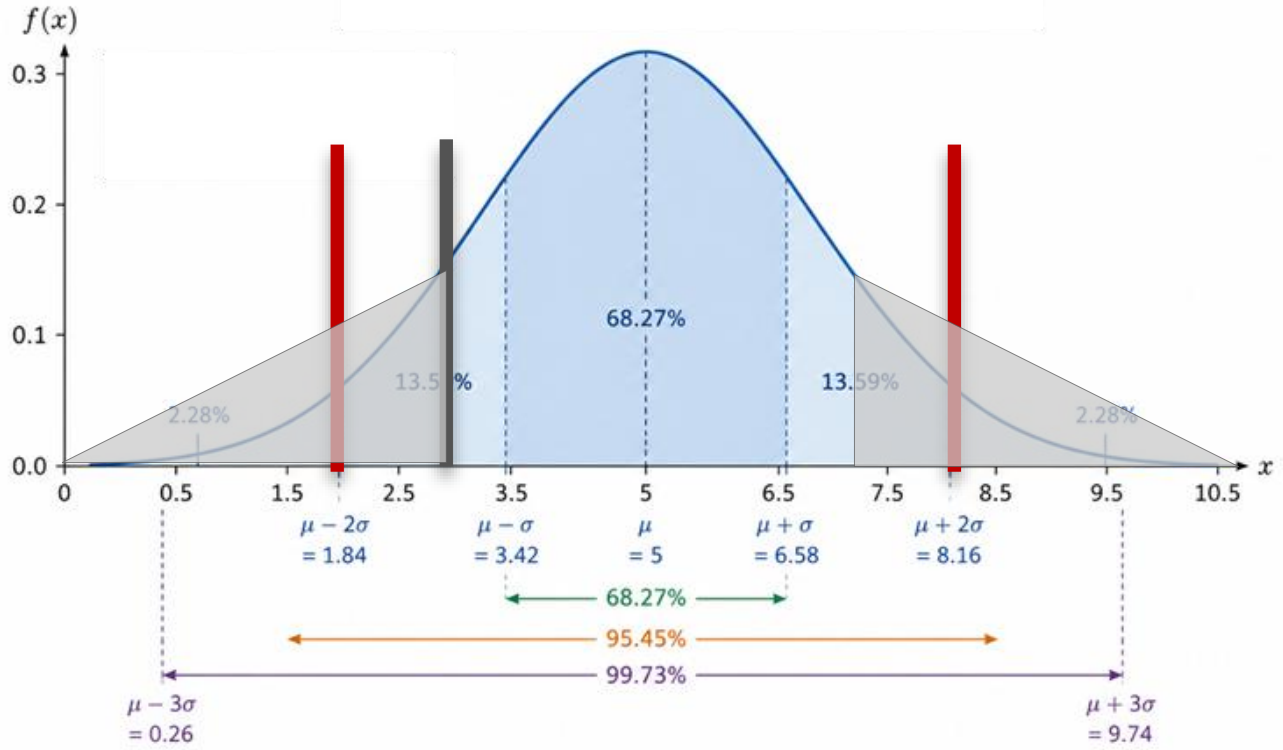
This is called p-value

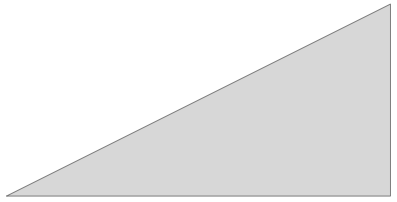
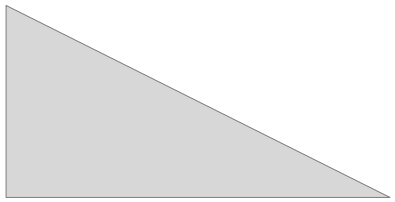
If p-value is low (0.001) we can reject H_0
If p-value is high (0.8) we cannot reject H_0

it is a bit different that say we accept H_0. In general we do not know how probable it is to observe the data if H_0 is false, maybe it is probable anyway!

*This is not **p(H_0 is not true | data)**!!!*

P-value



P-value =  + 

Hypothesis Testing

Reject the null hypothesis if p-value < T
It means we do have enough evidence to reject the hypotesis

T is some threshold chosen **a priori** as the desired **significance level** of the test (e.g., T=0.05 or T=0.01)

T = 0.05 → there is less than 5% of probability to observe the data given H0 is true

Remember: This means that 5% of the times I will reject H0 even if it was legit

Hypothesis Testing

Not rejecting the null hypothesis does **not** mean that we have **proven it to be true**…

…We just do not have **enough evidence** to reject it!

Hypothesis Testing

Two-tailed vs. **One-tailed** tests

Example 1:
A company produced items with an average time of 100s.
A new procedure reduces the manufacturing time to 95s [it is 5 seconds faster than the old one]
The manager analyzes a sample of the production of 100 items. The average production time is 93 s. Is there sufficient evidence to support the new procedure? i.e., to conclude that the average manufacturing time is less than 95s?

In general, H0 is not the hypothesis we have to prove. We want to reject H0 (and accept H1)

$H_0 : \mu > 95$ the avg. production time becomes > 95; the procedure is NOT effective

$H_1 : \mu <= 95$ the avg. production time becomes <= 95; the procedure is effective



This is **one-tail test**

Hypothesis Testing

Two-tailed vs. **One-tailed** tests

Example 2

A company manufactures cylindrical metal components with a target diameter of **5 mm**. The quality-control engineer wants to verify whether the manufacturing process is still correctly centered on the target value. A sample of produced components is measured.

- H_0 : average diameter is 5 mm
- H_1 : average diameter is not 5 mm

This is **two-tail test**



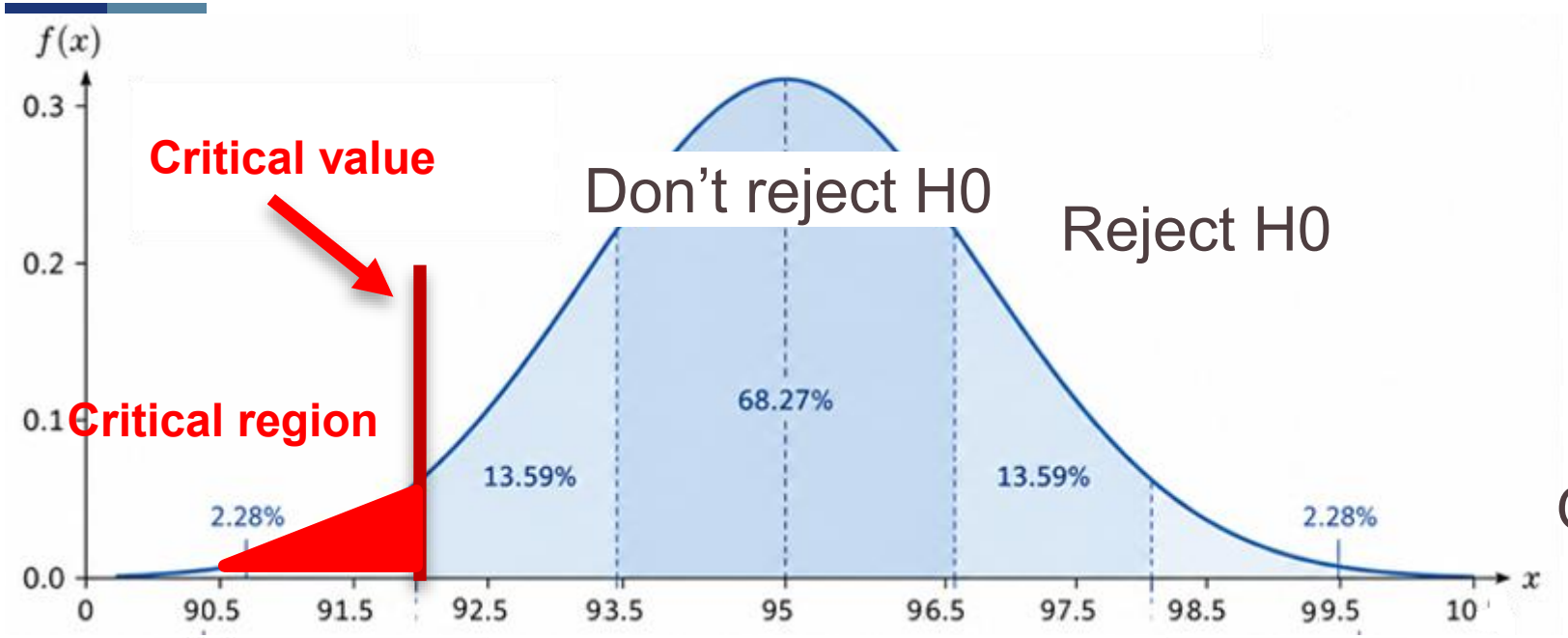
Hypothesis Testing

Two-tailed vs. **One-tailed** tests

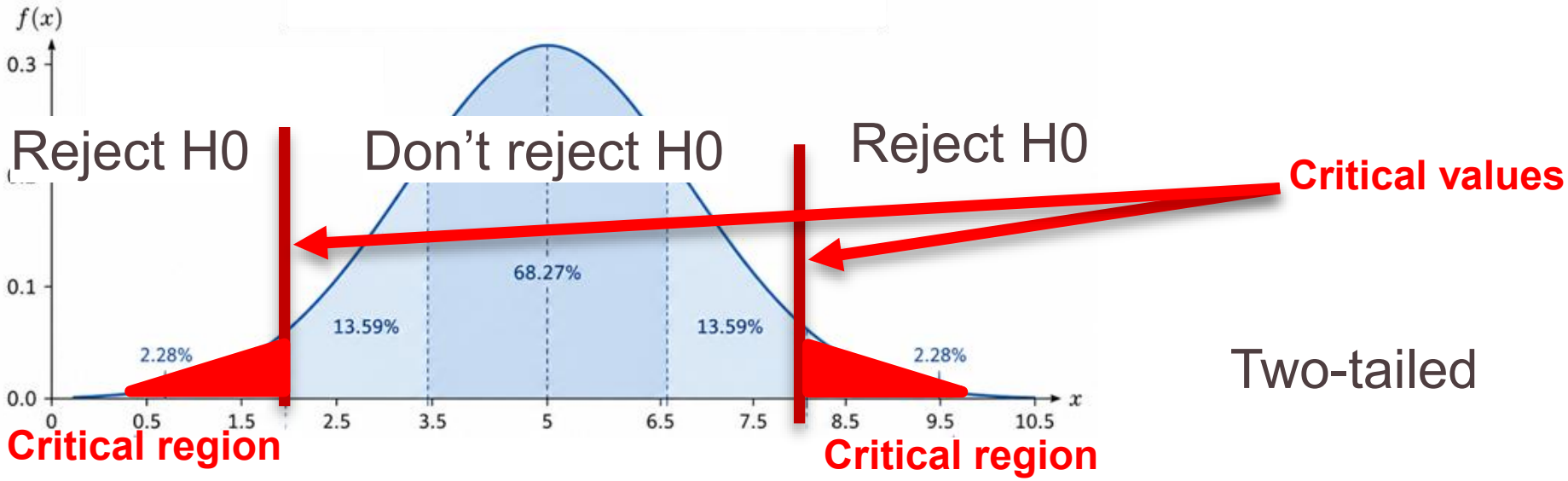
Example 1: “greater than” vs. “less than”
Example 2: “equal to” vs. “different from”

Example 1 should be treated with a **one-tailed** test
Example 2 should be treated with a **two-tailed** test

Hypothesis Testing



One-tailed



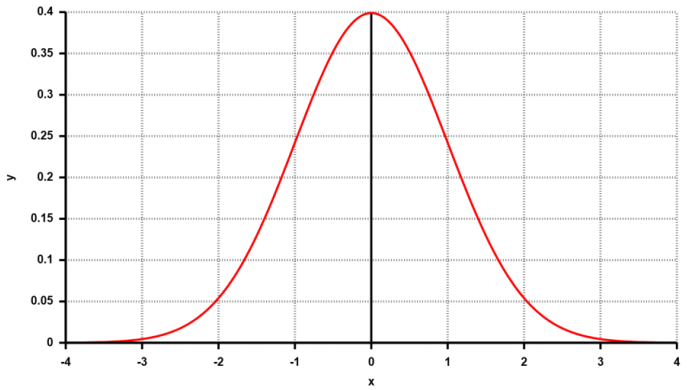
Two-tailed

Central Limit Theorem

Central Limit Theorem

Given **any** random variable X with **mean μ** and standard **deviation σ**, the sample mean of X follows the **normal distribution** (named Z) if the sample size is **sufficiently large**

$$\bar{Z} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \rightarrow N(0,1)$$

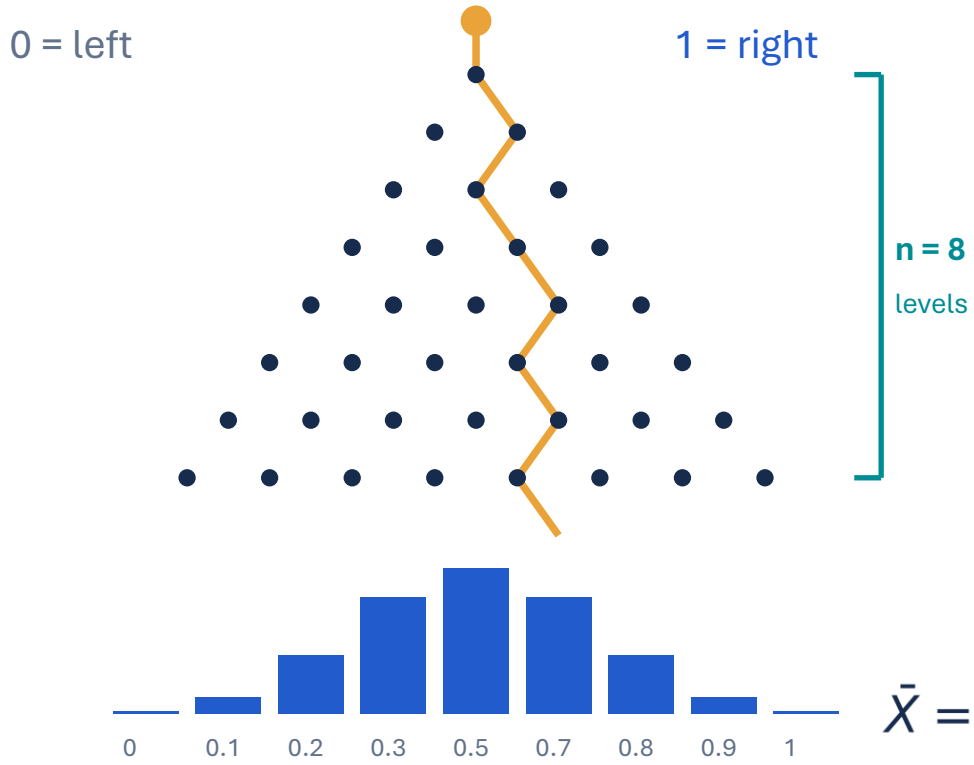


$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Galton board and the central limit theorem

Each ball sums n independent random choices: right or left, each with probability ½.

ONE BALL = ONE SAMPLE



Bin = number of right turns

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

THE MEAN OF THE CHOICES

X_i = 0 or 1, each with probability ½

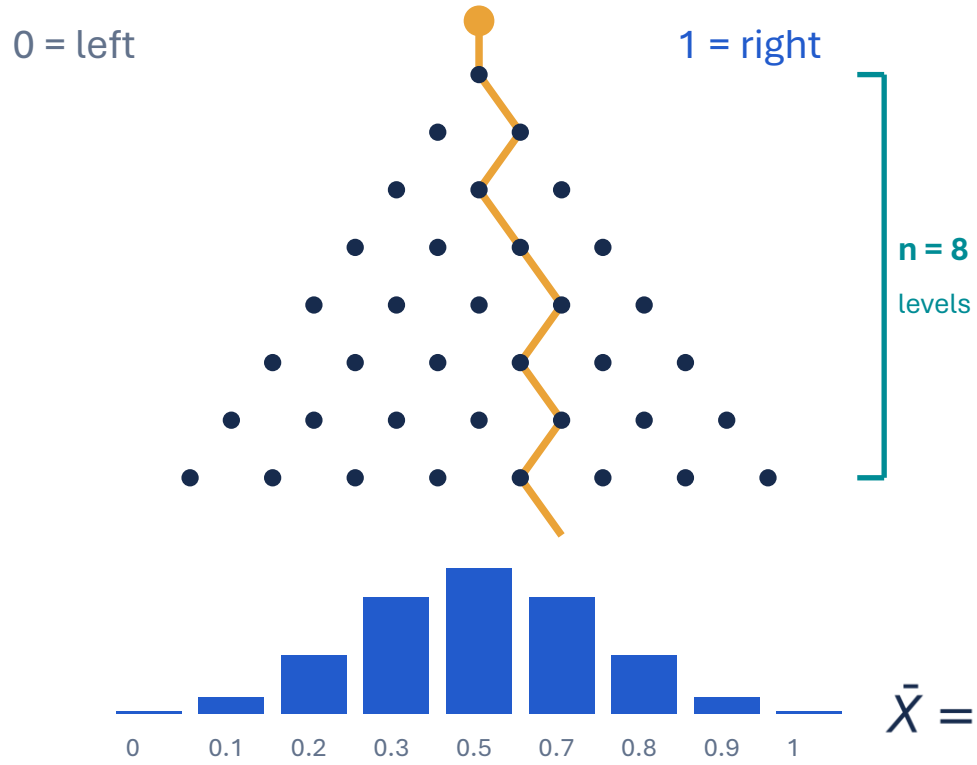
$\mu = E[X_i] = ½$ • $\sigma = ½$

Mean and standard deviation of a single choice.

$\bar{X}$ = proportion of right turns made by one ball = where the ball will land

Galton board and the central limit theorem

ONE BALL = ONE SAMPLE



$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$



FOR LARGE n

$$Z = \frac{\sqrt{n}\,(\bar{X} - \mu)}{\sigma} \approx N(0,1)$$

More levels → closer to a normal shape. **More balls** → a clearer view of the distribution.

Hypothesis Testing – Example 1

Example 1:
A company produced items with an average time of 100s.
A new procedure reduces the manufacturing time to 95s with standard deviation of 10s.
[it is 5 seconds faster than the old one]

The manager analyzes a sample of the production of 100 items. The average production time is 93 s. Is there sufficient evidence to support the new procedure? i.e., to conclude that the average manufacturing time is less than 95s?



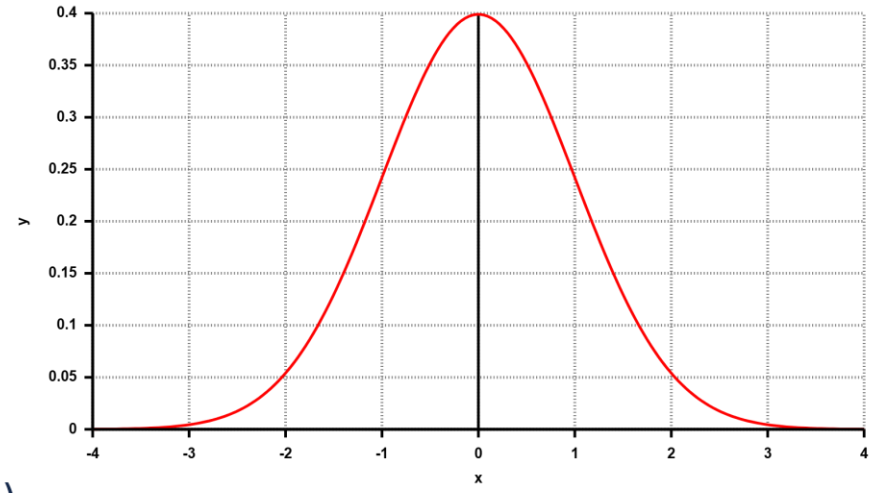
Central Limit Theorem

- **Analyze a batch of 100 items → average time = 93s**

Z-Transform

- **Z1 = √100 (93 – 95) / 10 = -2**

Central Limit Theorem

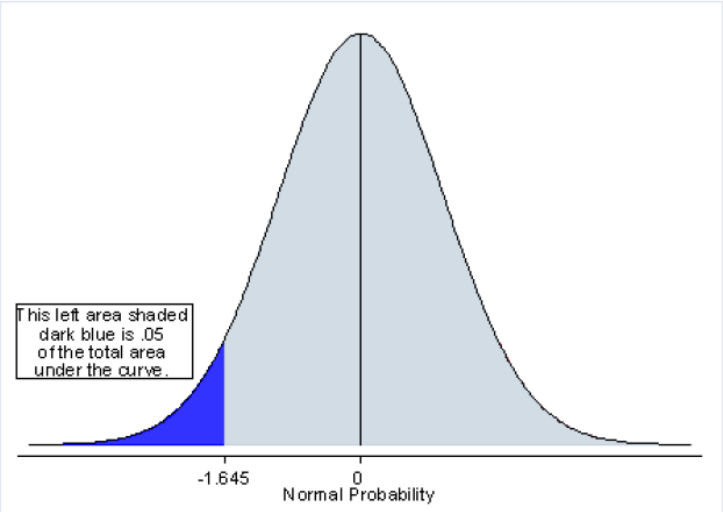


$$Z = \frac{\sqrt{n}\,(\bar{X} - \mu)}{\sigma} \approx N(0,1)$$

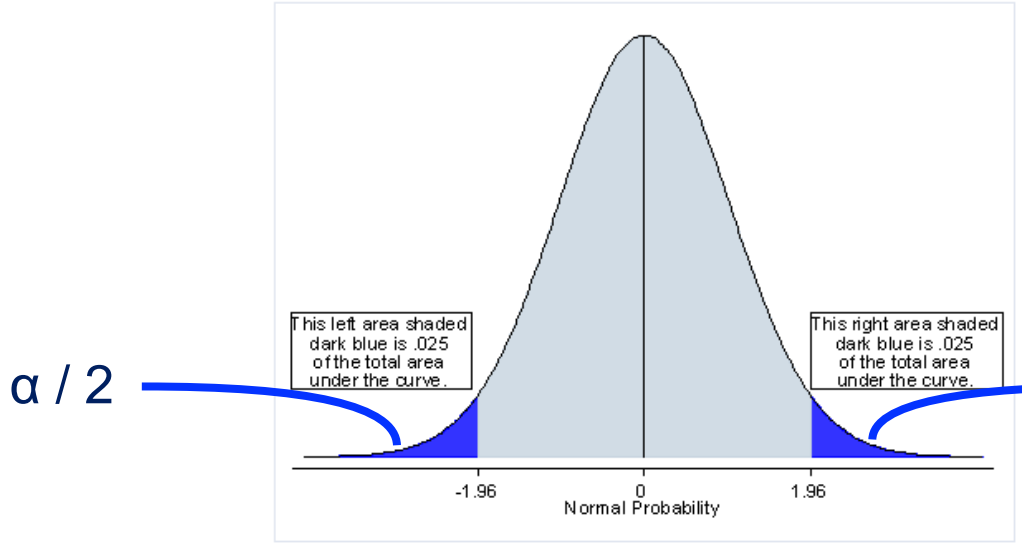
- The probability of getting 93s in a sample of 100 elements **from the general population not just from that sample** is computable from N(0,1)

Hypothesis Testing

Negative one-tailed test



Two-tailed test



$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

```
1  import scipy.stats as stats
2  p = 0.05  # significance level
3  z = stats.norm.ppf(p)
4  print(z)
5  p_value = stats.norm.cdf(z)
6  print(p)
7
```

✓ 0.0s

```
-1.6448536269514729
0.05
```

Hypothesis Testing – Example 1

We select H_0 as the hypothesis we want to falsify

$H_0 : \mu > 95$ the avg. production time becomes > 95; the procedure is NOT effective

$H_1 : \mu <= 95$ the avg. production time becomes <= 95; the procedure is effective

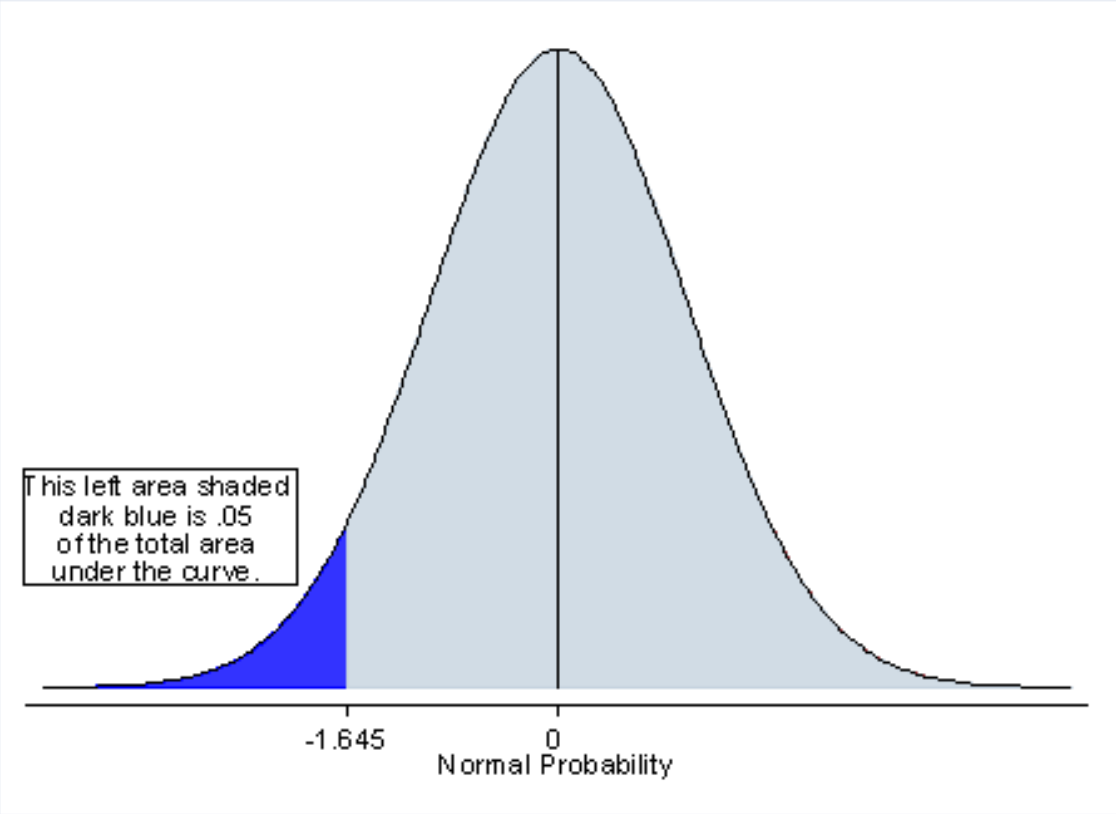
We fix in advance a 95% **confidence interval (1-α) and 5% significance (α)**
Compute statistics for the **mean with known std**

$$\bar{Z} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} = \sqrt{100} * (93 - 95) / 10 = -2$$

Does the statistic test fall **within** the 95% interval or **outside**?

One-tailed test!

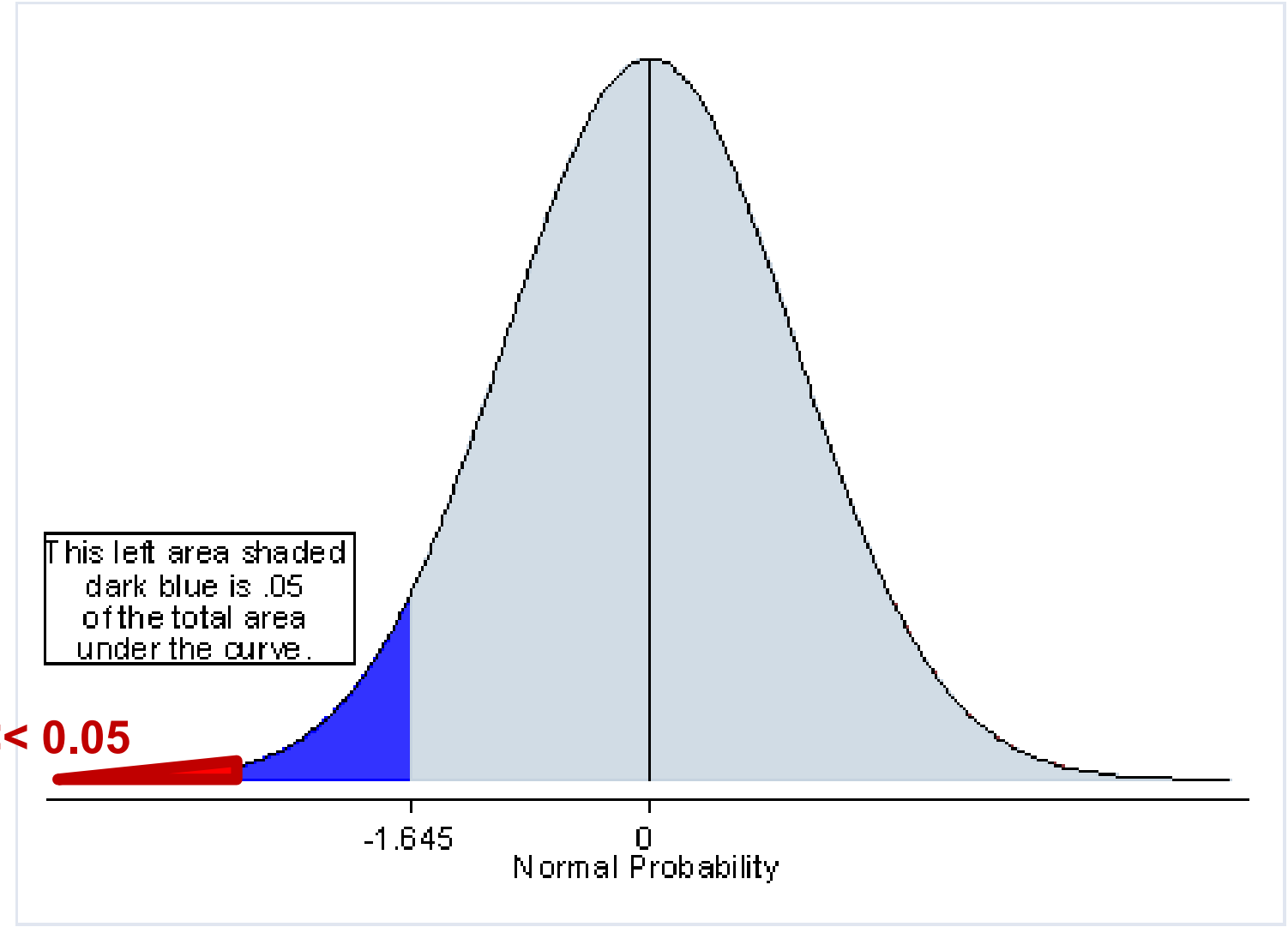
Hypothesis Testing – Example 1



Yes, it does!
-2 < -1.645

We reject H0 at 0.05 significance level, i.e., We reject that the average production time is greater than 95s. The procedure is effective!

Hypothesis Testing – Example 1



p-value << 0.05

The p-value is the area corresponding to the computed statistics

Hypothesis Testing – Example 2

The average weekly sales of a company is 1M
A new advertising campaign aims at increasing sales by 10%, leading to 1.1M and with a standard deviation of 50K.

The manager analyzes sales data over 10 weeks. The average weekly sales is 1.2M. Is there sufficient evidence to support the goodness of the campaign?



Hypothesis Testing – Esample 2

$H_0 : \mu <= 1.1M$

$H_1 : \mu > 1.1M$

One-tailed test!

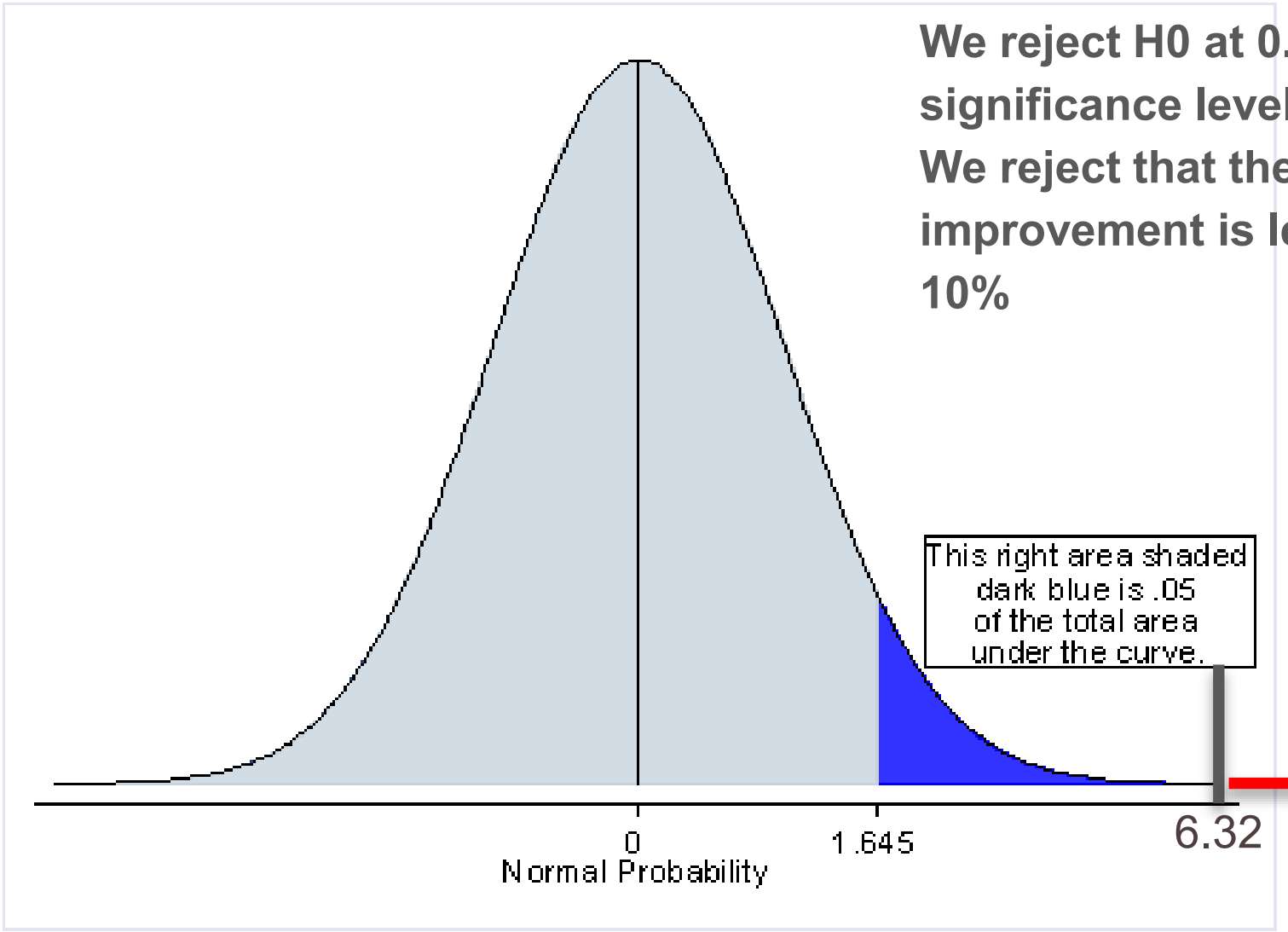
We fix in advance a 95% **confidence interval**

Compute statistics for the **mean with known std**

$$\bar{Z} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} = \sqrt{10} * (1.2 - 1.1) / 0.05 = 6.32$$

Does the statistic test fall **within** the 95% interval?

Hypothesis Testing – Example 2



We reject H0 at 0.05 significance level, i.e., We reject that the average improvement is less than 10%

p-value << 0.05

Hypothesis Testing – Example 3

A company manufactures cylindrical metal components with a target average diameter of **5 mm and standard deviation of 0.1 mm.**
The quality-control engineer wants to verify whether the manufacturing process is still correctly centered on the target value. A sample of produced components is measured.

The manager analyzes a sample of the production of 20 items. The average diameter is 5.01 mm.
Is there sufficient evidence to support his thesis?



Hypothesis Testing – Example 3

$H_0 : \mu = 5$ mm

$H_1 : \mu \; != 5$ mm

Two-tailed test!

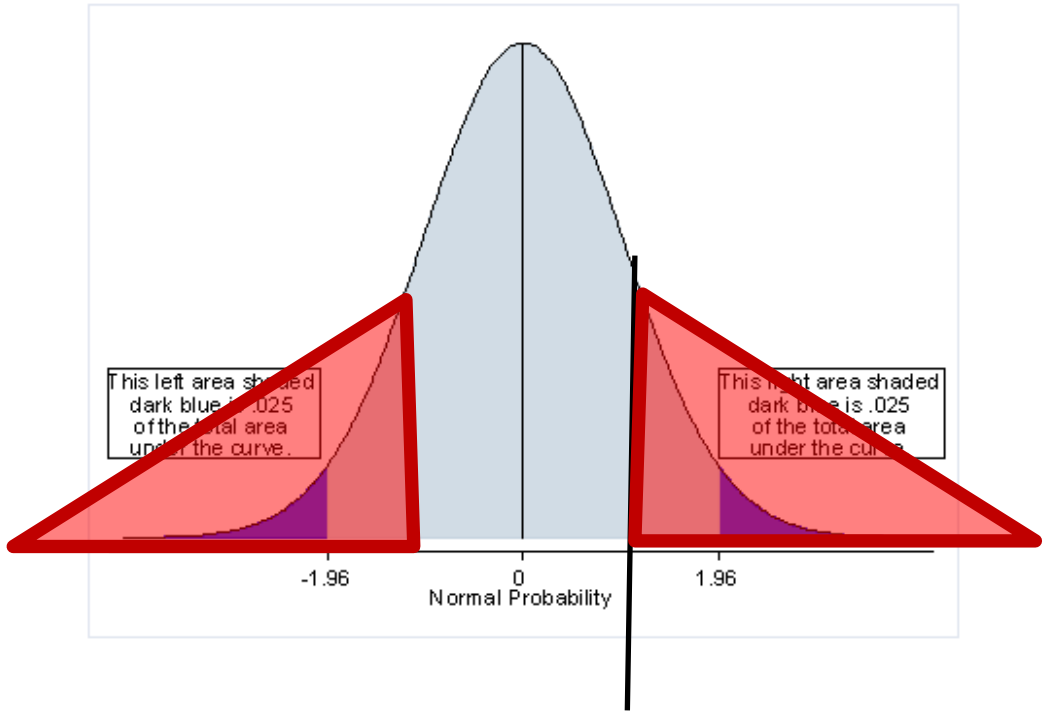
We fix in advance a 95% **confidence interval**

Compute statistics for the **mean with known std**

$$\bar{Z} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} = \sqrt{20} * (5.01 - 5) / 0.1 = 0.45$$

Does the statistic test fall **within** the 95% interval?

Hypothesis Testing – Example 3



p-value >> 0.05

We cannot reject H0 at 0.05 significance level, i.e.,
We cannot say that the process is not calibrated

Hypothesis Testing – Confidence Interval

It is also possible to compute a **confidence interval** for the mean, that is an interval estimate at a certain significance level:

$$(\bar{X} - Z\frac{\sigma}{\sqrt{n}}, \bar{X} + Z\frac{\sigma}{\sqrt{n}})$$

Given a significance level (e.g., 0.05) one finds the corresponding Z number (e.g., 1.645) and thus can compute the interval using σ and $\sqrt{n}$

Hypothesis Testing

Knowing the standard deviation in advance is a strong **assumption** that in practice often does **not** hold…

When the standard deviation is **unknown**, we can estimate it via the following formula:

$$s = \sqrt{\frac{\sum_i (X_i - \hat{X})^2}{n-1}}$$

Hypothesis Testing

Why dividing by n-1 instead of n?

If we take many **different** samples, and we average the standard deviation over all such samples, we would like to obtain an **estimate** of the standard deviation over the **whole population**…

…It turns out this happens **only** if we divide by n-1

```python
import numpy as np

sigmas_0 = []
sigmas_1 = []

population = np.random.normal(94, 10, 1000000)

for i in range(1000):
    sample = np.random.choice(population, size=100, replace=False)
    sigma_x_0 = np.std(sample, ddof=0)
    sigma_x_1 = np.std(sample, ddof=1)
    sigmas_0.append(sigma_x_0)
    sigmas_1.append(sigma_x_1)

print(f"Mean of sample std devs (ddof=0): {np.mean(sigmas_0):.6f}")
print(f"Mean of sample std devs (ddof=1): {np.mean(sigmas_1):.6f}")
```

✓ 17.6s

```
Mean of sample std devs (ddof=0): 9.916820
Mean of sample std devs (ddof=1): 9.966779
```

Hypothesis Testing

When the standard deviation is unknown **OR** the sample size is not large enough, it is **not safe** to use the Z distribution to approximate our data

What does **small sample size** mean?

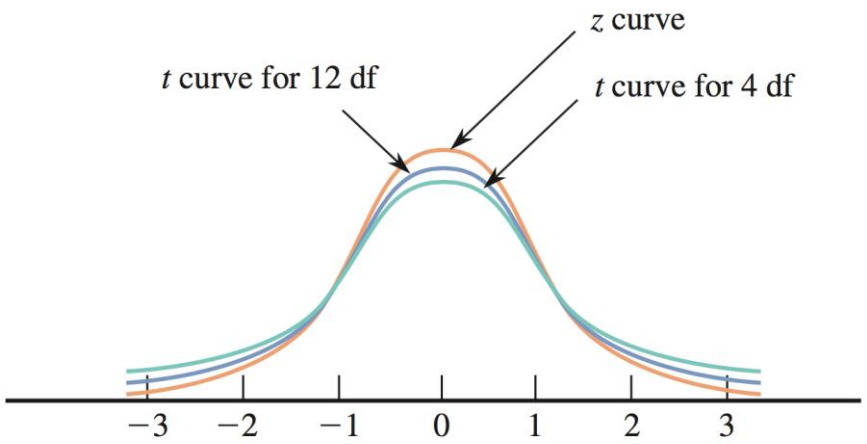
- In all statistics books, small means less than 30
- This is a **good rule of thumb** but…
- … Please do **not** take it as a gold standard!

Hypothesis Testing

In those cases use **Student's t distribution**

It has a **bell-like** shape similar to Z, but it depends on a certain parameter, named the **degrees of freedom**

It is actually a set of distributions



Hypothesis Testing

For the Student's t test:

- use **estimator** s instead of σ
- use n-1 as the **number of degrees of freedom**
- use thresholds for **Student's t distribution**

Hypothesis Testing – Example 1 (Revistited)

Example 1:
A company produced items with an average time of 100s.
A new procedure reduces the manufacturing time to 95s ~~with standard deviation of 10s.~~
[it is 5 seconds faster than the old one]

The manager analyzes a sample of the production of 15 items. The average production time is 93 s. **And an estimation of standard deviation s = 10 s.** Is there sufficient evidence to support the new procedure? i.e., to conclude that the average manufacturing time is less than 95s?



Hypothesis Testing

$H_0 : \mu > 95$
$H_1 : \mu <= 95$

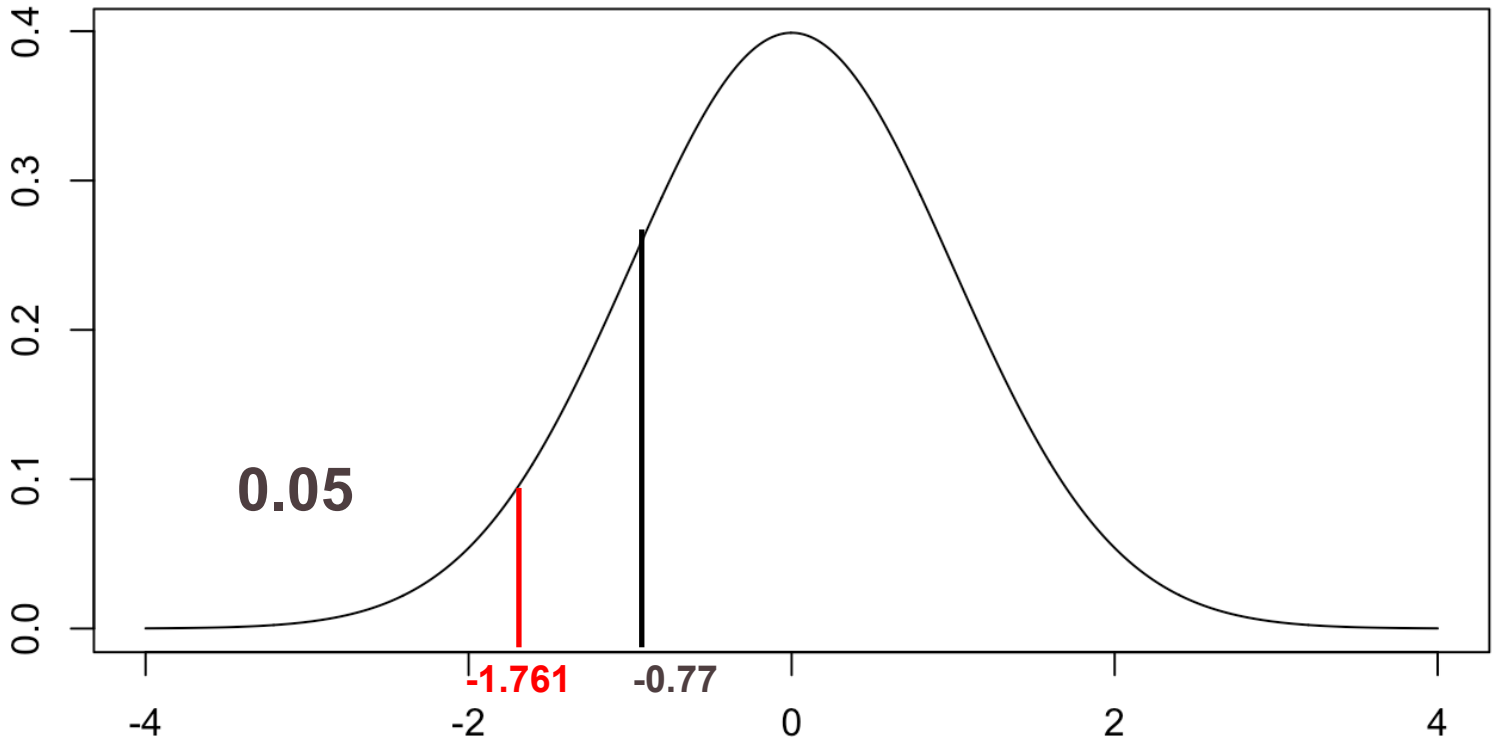
One-tailed test!

We fix in advance a 95% **confidence interval**
Compute statistics for the **mean with unknown std**

$$t = \frac{\sqrt{n}(\bar{X} - \mu)}{s} = \sqrt{15} * (93 - 95) / 10 = - 0.77$$

Does the statistic test fall **within** the 95% interval?
t-test with **14 degrees of freedom** (n-1)

Hypothesis Testing



No, it does not!
-1.761 < -0.77

We do not reject H_0 at 0.05 significance level

Hypothesis Testing – Example 4

What if we have to analyze a **discrete variable**?

This corresponds to a **categorical** contingency table

Example — Births and the lunar cycle

Lunar phase	# days	# births
New moon	24	7,680
Waxing crescent	152	48,442
First quarter	24	7,579
Waxing gibbous	149	47,814
Full moon	24	7,711
Waning gibbous	150	47,595
Last quarter	24	7,733
Waning crescent	152	48,230

American Journal of Obstetrics and Gynecology [2005]

Hypothesis Testing – Example 4

Example — Births and the lunar cycle

If there is **no effect** of lunar cycle on births, the number of births should be just **proportional** to the number of days. If it is not, then Moon has an effect.

Hypothesis Testing – Example 4

Example — Births and the lunar cycle

Lunar phase	# Days	# Births	Proportion	Expected # Births
New moon	24	7,680	0.0343	7,641.49
Waxing crescent	152	48,442	0.2175	48,455.52
First quarter	24	7,579	0.0343	7,641.49
Waxing gibbous	149	47,814	0.2132	47,497.55
Full moon	24	7,711	0.0343	7,641.49
Waning gibbous	150	47,595	0.2146	47,809.45
Last quarter	24	7,733	0.0343	7,641.49
Waning crescent	152	48,230	0.2175	48,455.52
TOTAL	699	222,784		

Hypothesis Testing – Example 4

Example — Births and the lunar cycle

In this case the **null hypothesis** is:

- expected = observed counts for category 1
- expected = observed counts for category 2
- …
- expected = observed counts for category K

None of the expected counts has to be **too small**

Hypothesis Testing – Example 4

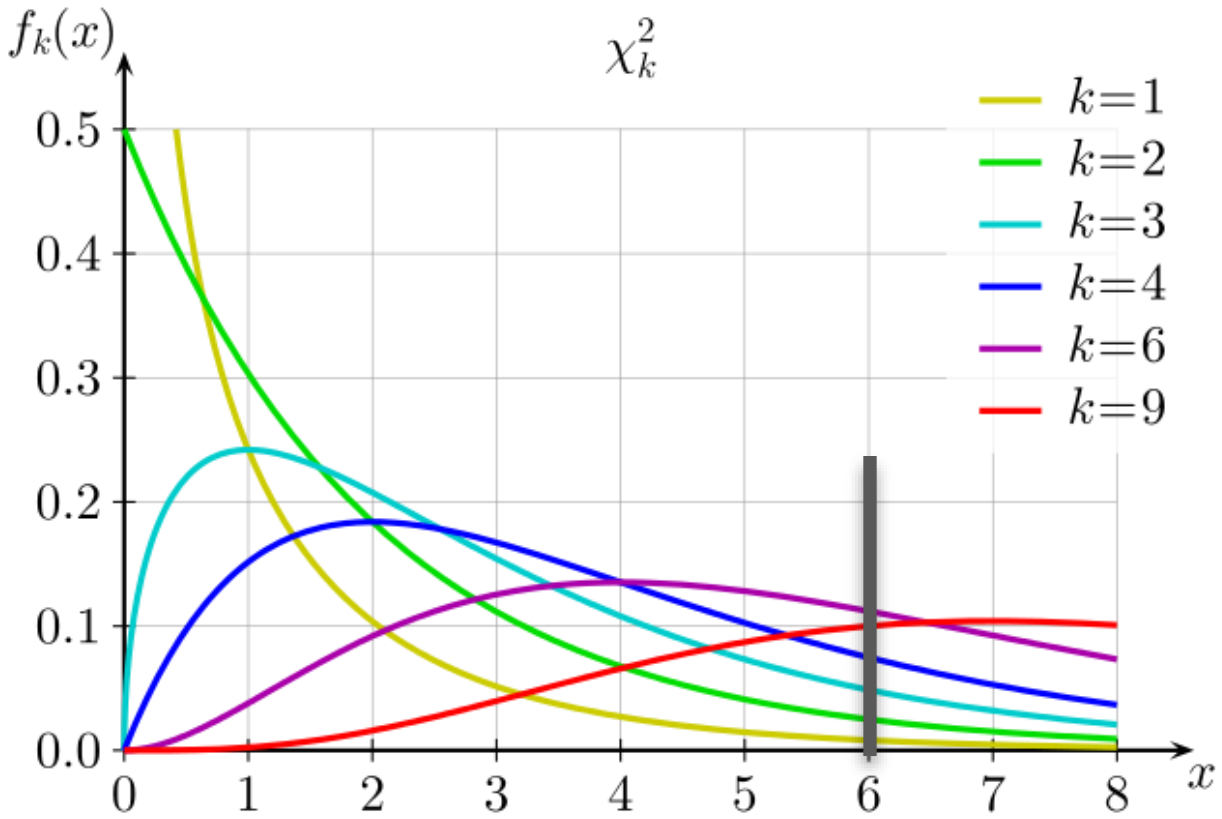
Example — Births and the lunar cycle

When a null hypothesis that specifies K category **proportions** is true, the goodness-of-fit statistic has approximately a χ^2 distribution with K-1 dof

goodness-of-fit statistics, where and are the observed and expected counts for category c, respectively

$$\chi^2 = \sum_c \frac{(o_c - e_c)^2}{o_c}$$

Hypothesis Testing – Example 4



0 or small variations between actual VS. expected counts

Large variations between actual VS. expected counts

χ^2 Tests are one-tail (the right tail: I want to check for large differences, becuase typically H0 is no relationship)

Hypothesis Testing – Example 4

Example — Births and the lunar cycle

Degrees of freedom = 8 - 1 = 7
Significance level = 0.05
Threshold = 14.07 [this is computed from the χ^2 distribution]
χ^2 = 0.194 + 0.004 + 0.511 + 2.108 + 0.632 + 0.962 +1.096 + 1.050 = **6.557**

Since **6.557 < 14.07** we cannot reject the null hypothesis, thus **there is not sufficient evidence to conclude that birth date and lunar cycle are related**

Hypothesis Testing – Example 5

- Hypothesis testing for Pearson correlation is very useful in data science
- It requires specific statistics

A manufacturing company wants to determine whether higher machine temperature is associated with longer production times.
The company collects $n = 30$ paired observations:

- machine temperature;
- production time for the corresponding item.

The sample Pearson correlation is: $r = 0.40$.
The manager wants to test, at the 5% significance level, whether there is a positive linear correlation in the population.

Hypothesis Testing – Example 5

- H0: r = 0 ; H1: r > 0
- **Find z value with Fisher's transformation (see code)**
- z = 2.2
- Compute the p_value as in the z-test case
- p_value = 0.014
- 0.014 < 0.05
- Therefore we reject the null hypotesis of **absence** of correlation between the variables

Why not a t-test?
t-tests are for comparing means
Pearson correlation has different properties than a mean
For one, it is confined between -1 and 1,
while the t-distribution goes from –inf, inf

Equivalent of central limit theorem. This quantity has a N(0,1) distribution!

Hypothesis Testing

There exist **many more** statistical tests!

We will see a few more examples in the part of the course dedicated to **machine learning**
Some of such statistical tests are in fact used to **compare** two distinct classifiers

Hypothesis Testing

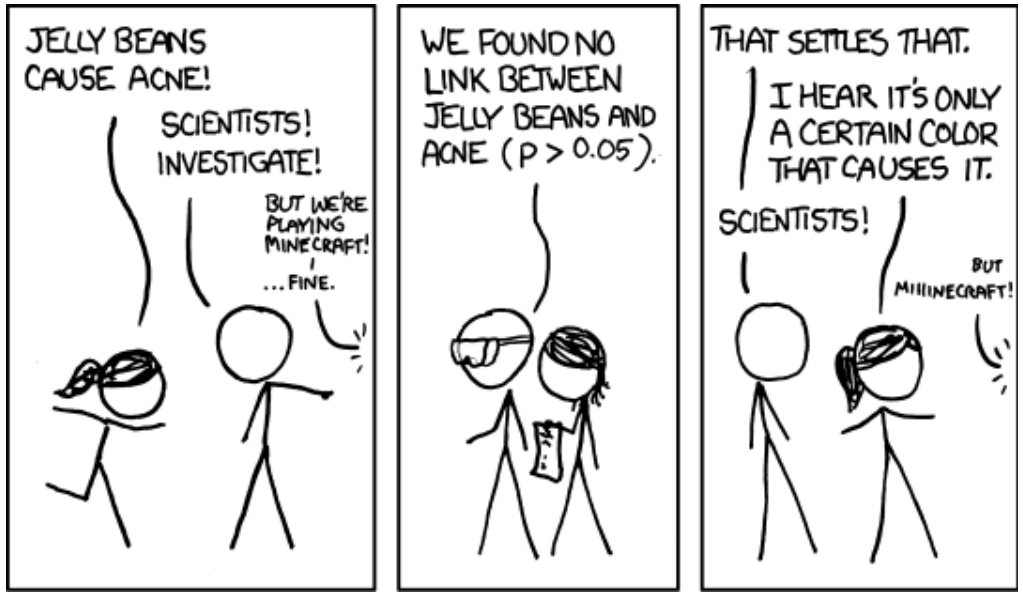
Statistical vs. practical significance

Not necessarily a **statistically significant** result has some **practical** consequence… It really depends on the **context**!

For example bacuse the the observed difference is too small

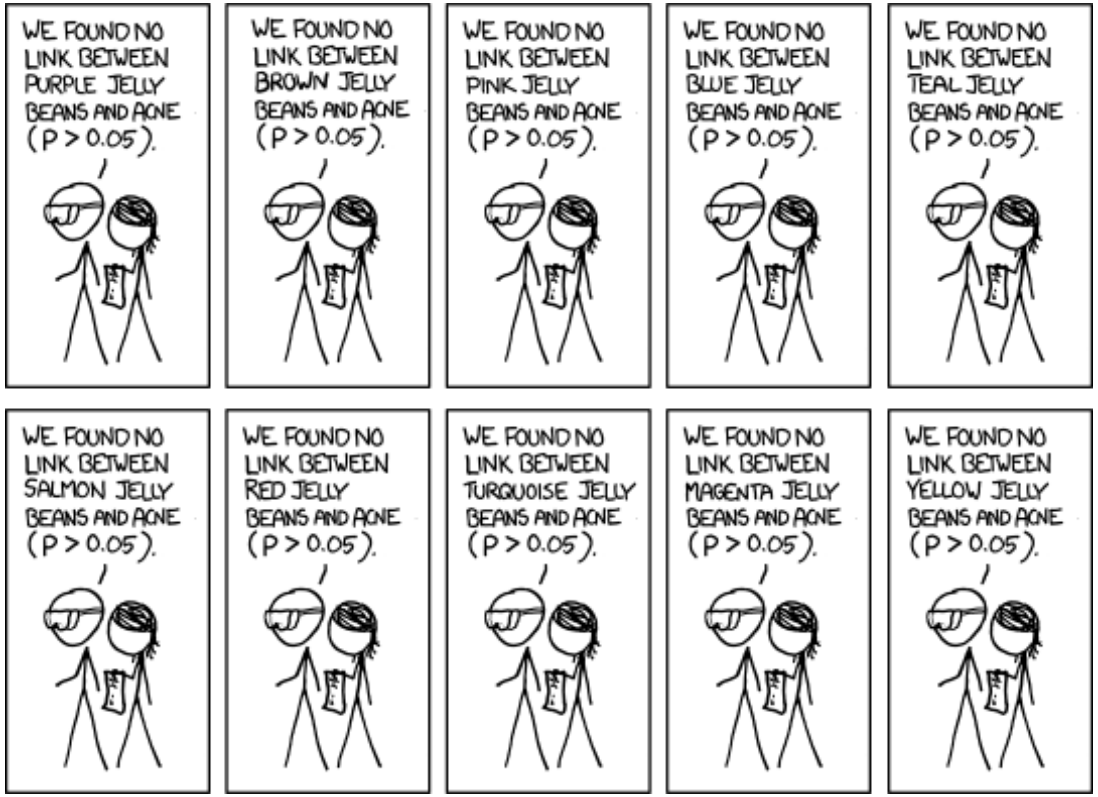
Hypothesis Testing

Beware of repeated experiments and p-value… (1/4)



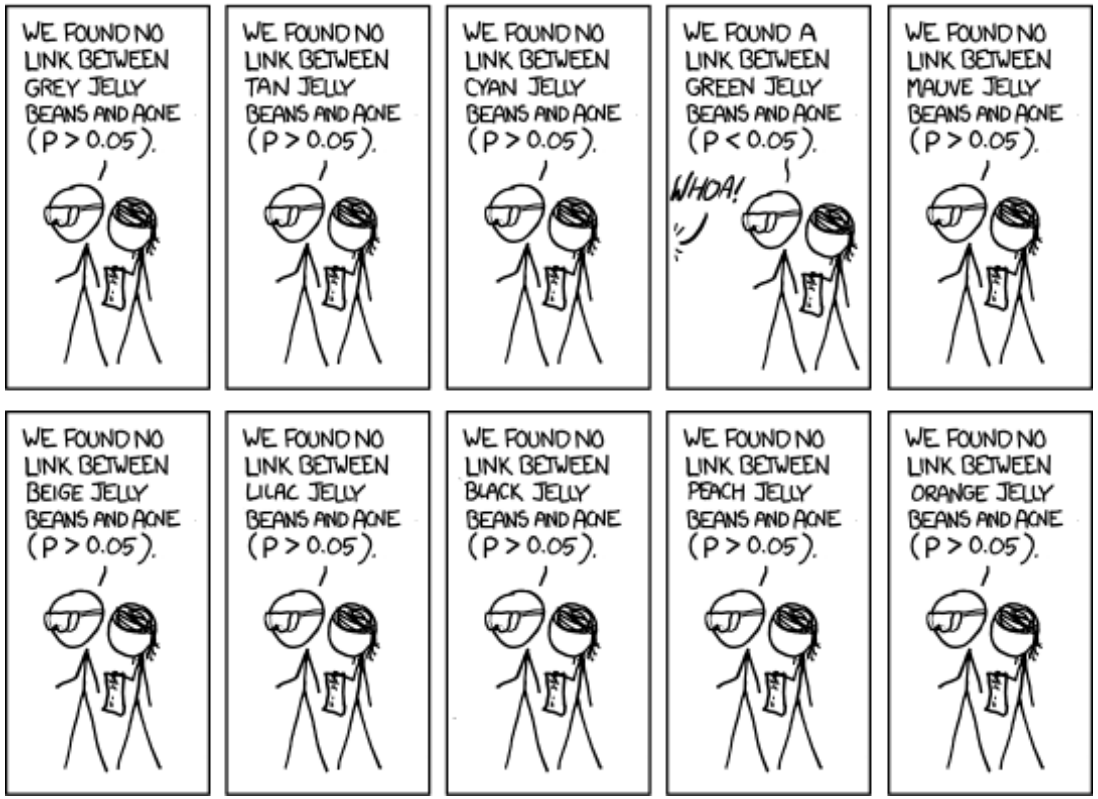
Hypothesis Testing

Beware of repeated experiments and p-value… (2/4)



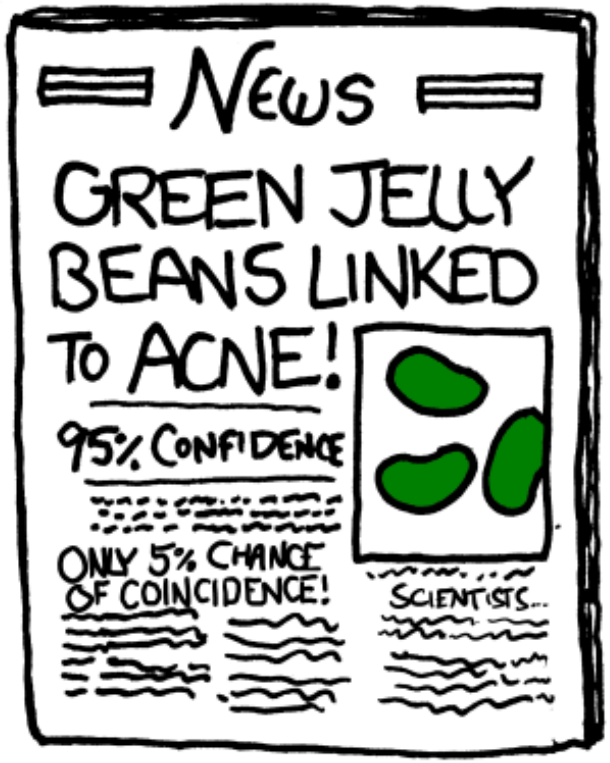
Hypothesis Testing

Beware of repeated experiments and p-value… (3/4)



Hypothesis Testing

Beware of repeated experiments and p-value… (4/4)



Bonferroni
When dealing with multiple tests (m tests)

$\alpha = 0.05 / m$

to keep overall false positive risk equal to 0.05

Hypothesis Testing - Limitations

- Limitations of **frequentist** approaches
- Strongly dependent on the **sample size**
- The p-value **is not** the probability of H0 being true
- Confidence interval **is not** a probability distribution
- Are some experiments **really** reproducible?

Statistical Power

	H_0 true	H_1 true
Reject H_1	$1-\alpha$	β
Reject H_0	α	$1-\beta$ ← POWER

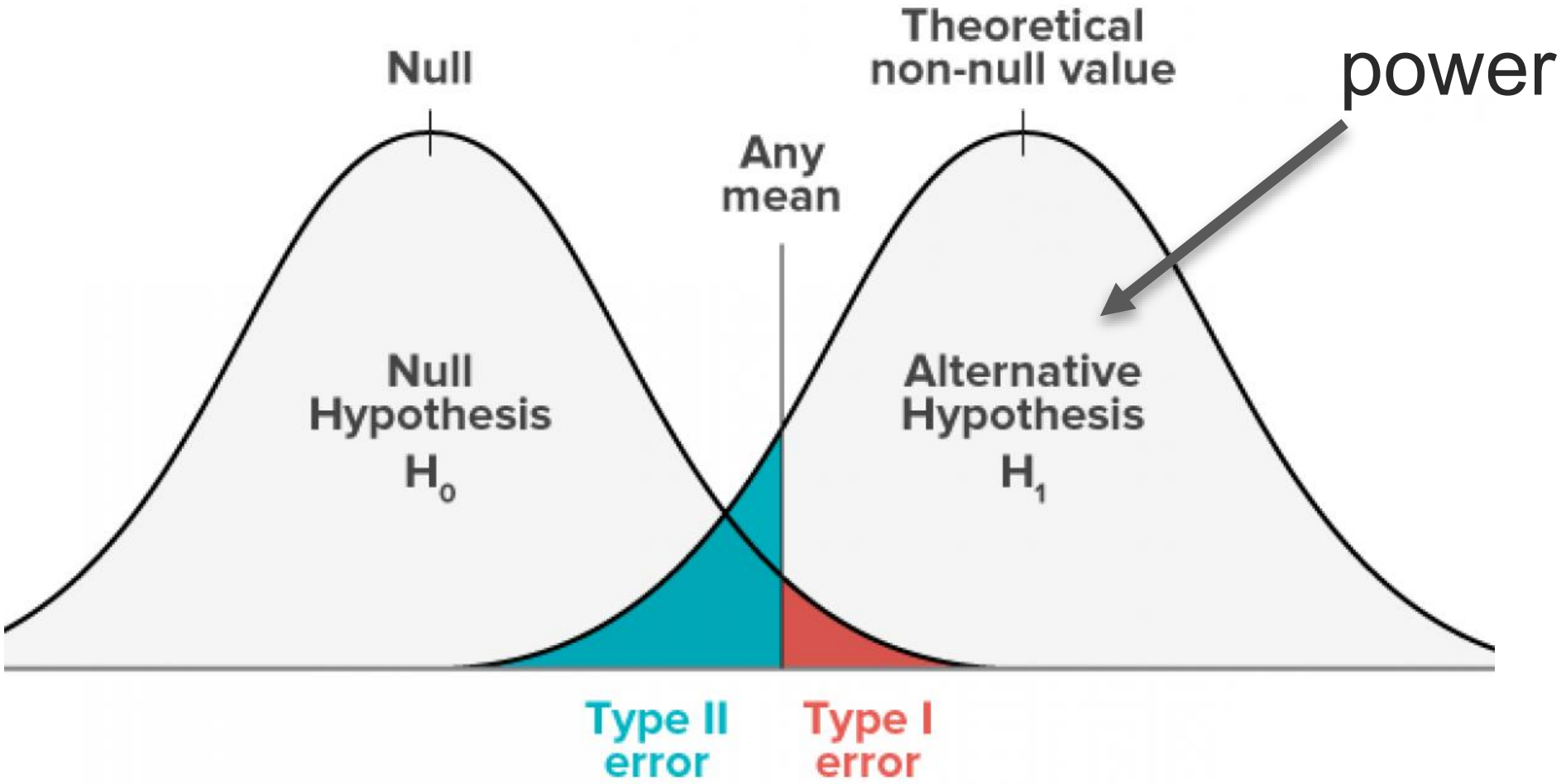
SIZE

- The **power of the test (1-β)** is the probability of rejecting H_0 in the case that H_1 is true (or H_0 is false)

- A statistical power of 80% means:
  - If the effect specified in the analysis truly exists in the population, repeating the study many times would produce a statistically significant result in approximately 80% of those studies.
  - In the remaining 20% of cases, the study would fail to detect the effect even though it exists. This is known as a Type II error and is denoted by β.

- Importantly, statistical power does not mean that there is an 80% probability that the research hypothesis is true. It describes the probability that a statistical test will detect an effect of a specified size when that effect genuinely exists.

Power


Null
Theoretical
non-null value
power
Any
mean
Null
Hypothesis
H_0
Alternative
Hypothesis
H_1
Type II
error
Type I
error

The main factors affecting power

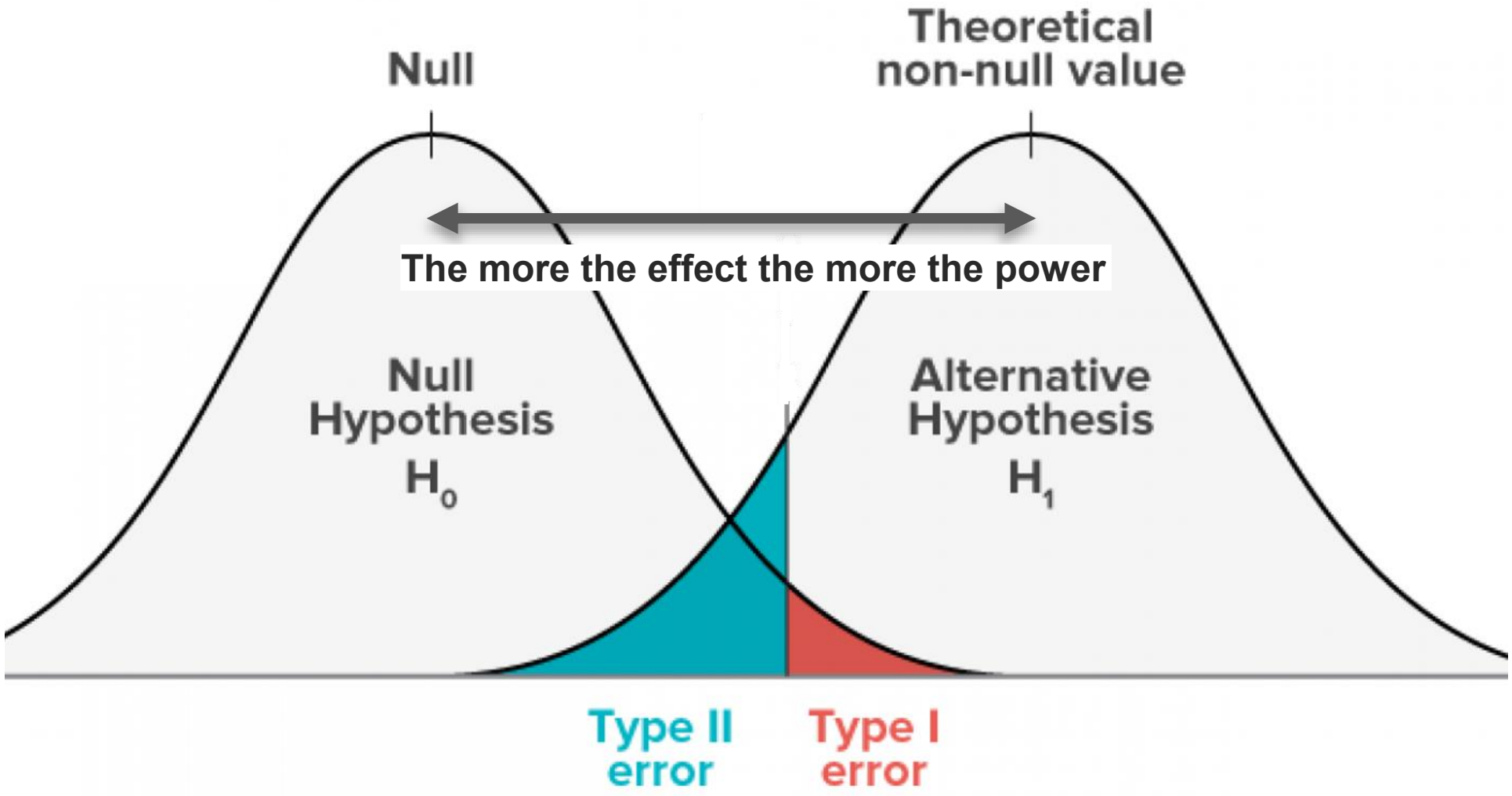
Factor	Meaning	Effect on power
Sample size N	Number of observations or participants	Larger sample → higher power
Effect size	Strength of the relationship or difference	Larger effect → higher power
Significance level α	Accepted probability of a false positive	Larger α→ higher power

In a classic power analysis, researchers normally specify:

1. The smallest effect they want to be able to detect (e.g., a procedure usually takes 100s if there is a change and now it takes 95s, I want to be able to measure it)
2. The significance level, usually $\alpha = 0.05$
3. The desired power, usually 80% or 90%
4. The statistical test that will be used

These inputs are then used to calculate the required sample size.

Power and Effect Size



Power Testing – Example 1

Example 1:
The average time to manufacture an item if 100 s, with a population standard deviation of 10 s.
A company invents a special manufacturing procedure to produce items **faster.**

- The manager will conduct a one-sided z-test at the 5% significance level:
- $H_0: \mu \geq 100$ versus $H_1: \mu < 100$.

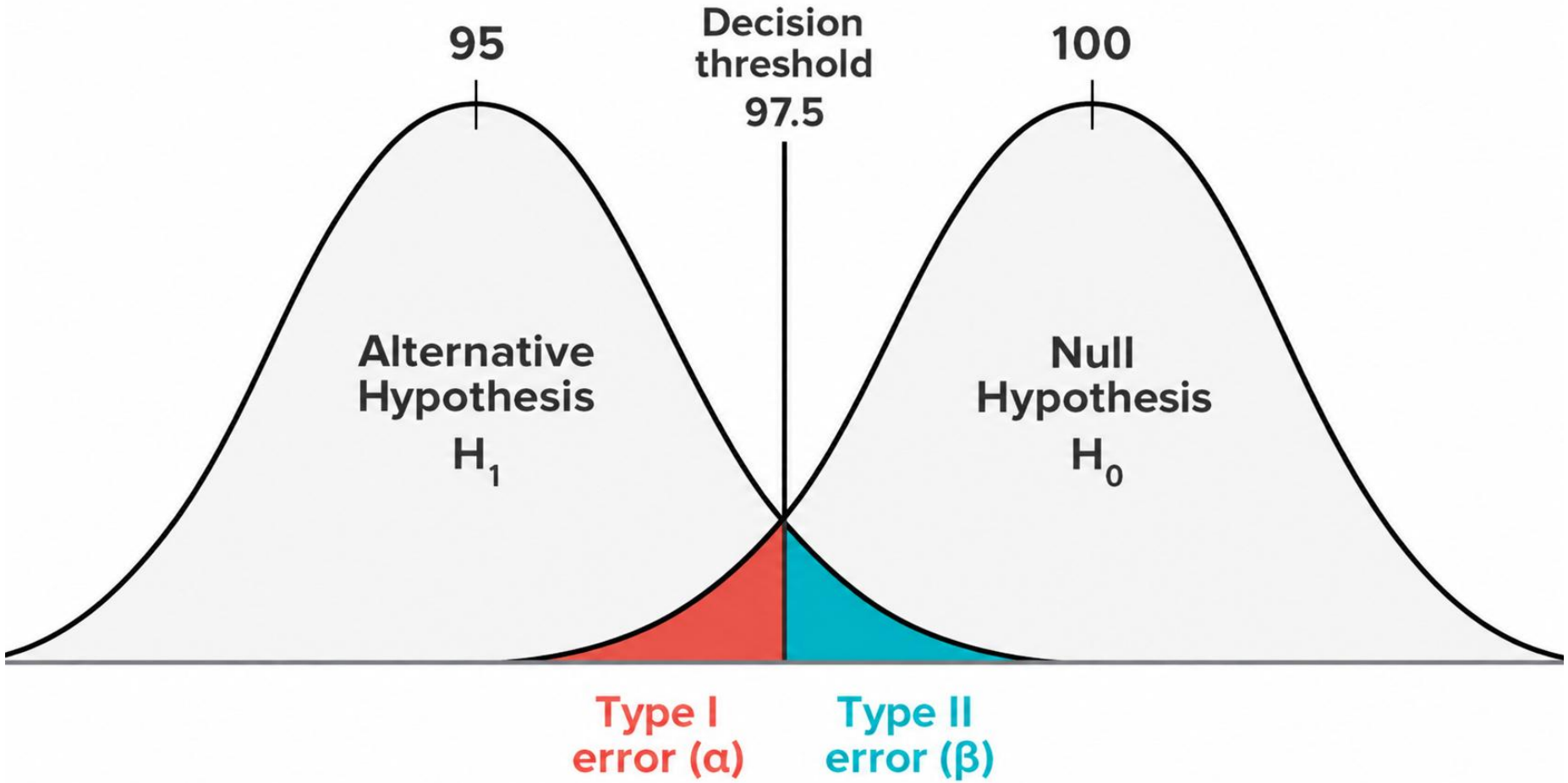
The manager wants the test to have 90% power if the true average manufacturing time under the new procedure is 95 seconds—that is, when the procedure reduces the average time by 5 seconds.

If the new procedure truly reduces the mean manufacturing time from 100 to 95 seconds, the test should produce a statistically significant result in 90% of repeated experiments.

How many items should the manager sample?



Power Testing – Example 1



Power Test – Example 2

- In a correlation study, How many participants are required to have an 80% probability of obtaining a significant result if the true correlation is 0.30?

- Significant result means:
- H0: r=0 versus H1: r !=0